A Hundred Years of Business Cycles and the Phillips Curve

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Abstract

US business cycles since 1900, examined through a multivariate trend-cycle model consistent with a New Keynesian economy, reveal robust reduced-form regularities between slack, prices and unemployment. Once long-term trends are accounted for, a single common cycle underlies output, unemployment, prices, inflation expectations and the policy rate, and behaves as a demand cycle. The slope of inflation on unemployment is -1.08 over the century, against -0.40 in the raw data, and around -0.80 in postwar subsamples. Okun's law, connecting unemployment and output, is an equally robust relation in the data.

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Business cycles are a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions [...] This sequence of changes is recurrent but not periodic. In duration, business cycles vary from more than one year to ten or twelve years. (Burns and Mitchell, 1946)

When the demand for a commodity or service is high relatively to the supply of it we expect the price to rise, the rate of rise being greater the greater the excess demand. Conversely when the demand is low relatively to the supply we expect the price to fall, the rate of fall being greater the greater the deficiency of demand. It seems plausible that this principle should operate as one of the factors determining the rate of change of money wage rates. (Phillips, 1958)

1 Introduction

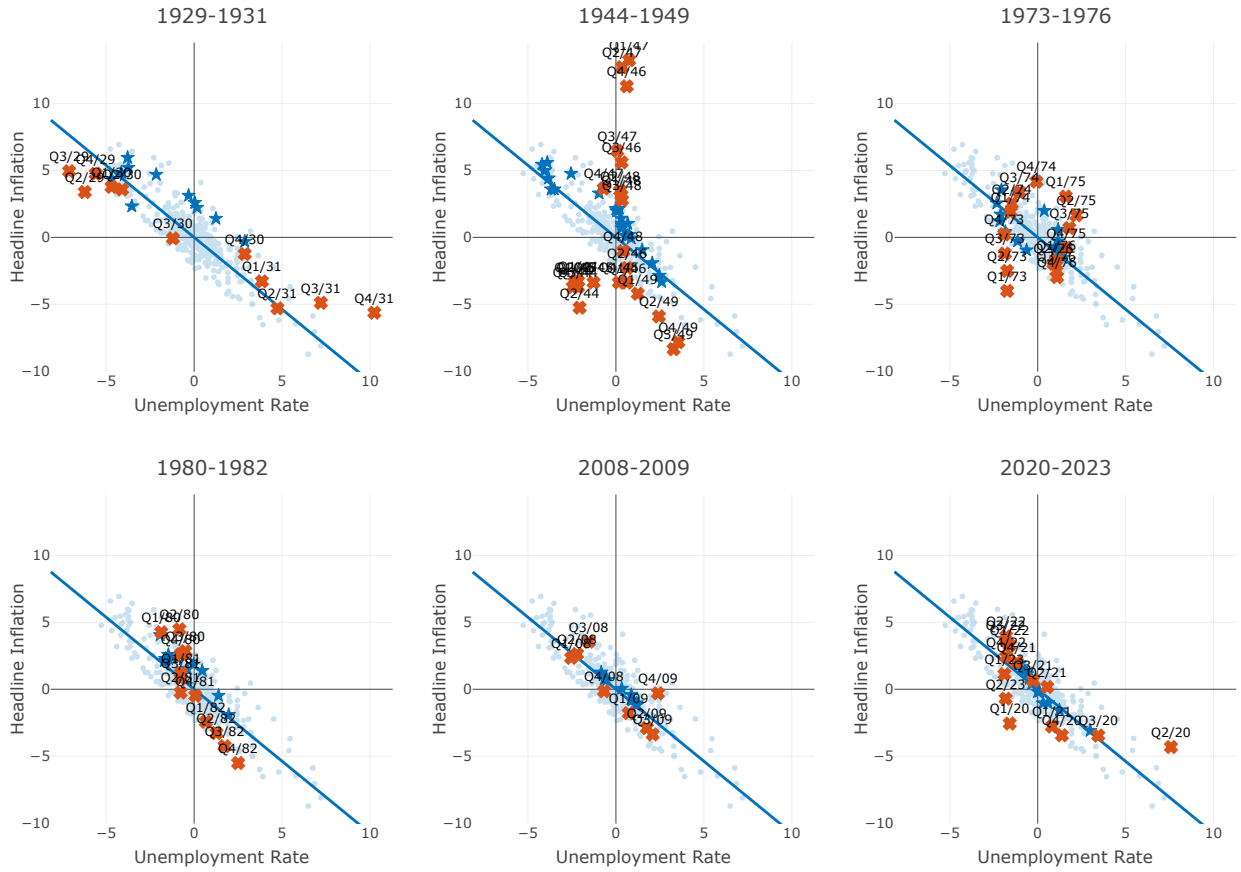
Almost seventy years after Phillips (1958) documented a negative correlation between economic slack and wage inflation in United Kingdom data, that correlation remains the most contested regularity in macroeconomics. Early work confirmed it elsewhere with simple reduced-form regressions of price or wage inflation on unemployment or the output gap. From the 1990s, a microfounded structural Phillips curve became the central building block of the New Keynesian model. Paradoxically, as its theoretical foundations waxed, the empirical relationship waned. The curve has since been declared alive and dead, killed by policy or by luck, steep or flat and, after the post-pandemic surge, non-linear or unstable. Papers on the subject number in the thousands.[1]

The aggregate evidence has settled neither question: whether the correlation is there at all, and, if it is, how strong it is. The forecasting literature reports poor out-of-sample performance: in an evaluation spanning many specifications, Stock and Watson (2009) find that forecasts based on Gordon (1990) perform in some subsamples but generally fail to improve on univariate models of inflation alone. The more model-based literature, aiming at the structural parameters of the New Keynesian equation, has run into weak identification; Mavroeidis et al. (2014) conclude that aggregate time series have been pushed about as far as they can go, and that progress requires new identification strategies and new data.

Figure 1 suggests why this might be the case. The orange crosses plot demeaned CPI inflation against the demeaned unemployment rate in six episodes of high inflation or deep recession. The Great Depression (1929–31), the Volcker recession (1980–82) and the Global Financial Crisis (2008–09) – all associated with monetary contraction, financial disruption, or both – trace a steeply negative relationship. The postwar inflation (1944–49), the Great Stagflation (1973–76) and the pandemic inflation (2020–23) trace an almost vertical one, reflecting a larger role for supply disturbances such as the removal of price controls, shortages and oil shocks. The raw

[1] We do not aim to review this literature. For recent surveys of the NKPC see Mavroeidis et al. (2014) and Furlanetto and Lepetit (2025); for Phillips-curve-based forecasting, Stock and Watson (2009).

Figure 1: Selected episodes of high inflation and recessions in the US.



Notes: Each panel plots demeaned CPI inflation against the demeaned unemployment rate (orange crosses), together with the model-based business cycle components of the same two series (blue dots) and the reduced-form Phillips curve they trace out (blue line). The model is estimated over the sample 1900–2019; the final panel applies that same model to data through 2023.

data are thus likely to conflate two distinct pairs of movements: fluctuations at business cycle frequency with shifts that permanently change levels; and, within the cycle, disturbances that move slack and prices along the curve with disturbances that push inflation off it.

This paper shows that a simple linear multivariate trend-cycle model recovers a steep, downward-sloping reduced-form Phillips curve at business cycle frequency, and that this curve is stable over more than a century of US data. We decompose seven quarterly US series – output, unemployment, headline and core inflation, inflation expectations of professional forecasters and consumers, and the policy rate – from 1900 to 2019 into a variable-specific non-stationary trend, a single common cyclical component, and a variable-specific idiosyncratic disturbance. The logic is straightforward: a flexible trend absorbs the low-frequency movements in prices, unemployment and output that have little to do with the cycle, while the idiosyncratic components absorb the transitory disturbances that obscure it. What remains is a common business cycle component that governs the comovement of nominal and real variables. The blue dots in Figure 1 are that component; unlike the raw data, they fall along a single downward-sloping line in every one of the six episodes.

Section 2 illustrates how a textbook New Keynesian economy augmented with stochastic trends admits a trend-cycle representation, with cycle loadings that are functions of the struc-

tural parameters.[2] This representation guides the empirical model of Section 3, which follows the trend-cycle tradition of Harvey (1985) and is estimated with Bayesian methods on more than a century of US quarterly data.

Despite being informed by a New Keynesian model, our approach is deliberately reduced-form, in the spirit of Burns and Mitchell (1946) and Phillips (1958), and the restrictions we impose are few. All seven series load on a single common cycle. Each variable has its own stochastic trend, with the exception of a single trend inflation that drives headline and core inflation, both survey expectations and the policy rate. The unobserved components are mutually orthogonal, an assumption we relax in robustness exercises by letting the innovations to the cycle and to the trends correlate. Under these restrictions the common business cycle component transmits to prices and price expectations through a reduced-form Phillips curve and to the labour market through Okun's law, while the trends are interpretable as potential output, the natural rate of unemployment and trend inflation. Survey expectations, in particular, help pin down the common price trend and, with it, the cycle.

The empirical analysis of Sections 4 and 5 delivers three key results. First, the common cycle accounts for a large share of US economic fluctuations, and it looks like a demand cycle. The model-based output gap lines up with the NBER recession dates and with the official CBO and Federal Reserve estimates, and matches the frequencies and amplitudes conventionally ascribed to business cycles. Along the cycle, contractions coincide with rising unemployment, lower interest rates and downward pressure on prices. These signs are not imposed: re-estimating the model under sign restrictions that make the demand interpretation explicit leaves the posteriors of the loadings essentially unchanged, since the unrestricted posteriors already place almost all their mass on the demand-consistent side. The common cycle also dominates the policy rate, available from 1954 onwards: its cyclical movements are largely accounted for by the common cycle, and the idiosyncratic, 'erratic' part is small and largely confined to the Volcker disinflation. This is consistent with the view that monetary policy has been largely systematic over the postwar period.

Second, the reduced-form Phillips curve is steep and stable. At business cycle frequency a given amount of slack maps into a specific and sizeable movement in inflation. The slope of inflation on unemployment is -1.08 over the full century, against -0.40 in the demeaned data, and it remains steeply negative when the model is re-estimated on each of two subperiods: -0.77 over 1946–84 and -0.86 over 1985–2019. A rolling-window exercise, re-estimating the full model on eighty-year windows from a first window ending in 1980 to a last ending in 2019, keeps the slope firmly negative throughout, within a range of roughly -0.5 to -1.1. There is some flattening from the windows ending in the early 2000s, which have shed the volatile early decades of the century and estimate the slope from progressively less cyclical variation, but in every window the 90% band of the slope excludes zero. Once its components at business cycle frequency are properly isolated, the reduced-form Phillips curve is alive and well.

[2] Theory thus disciplines the shape of the empirical model rather than imposing cross-equation constraints on its coefficients: in the empirical analysis the loadings on the common cycle are left free and unrestricted in sign, so that the data inform them.

Third, Okun's law is at least as stable as the Phillips curve. The same common cycle that ties slack to inflation also ties the business cycle component of GDP to that of unemployment. Over the full century our Okun's law coefficient is -0.417, essentially identical to the -0.421 of Ball et al. (2017). Re-estimating over the same two subperiods gives -0.33 in 1946–84 and -0.52 in 1985–2019. We thus confirm their conclusion that Okun's law remains one of the most stable relationships over time, and reinforce our reading of the common cycle as a robust, demand-like driver of joint fluctuations in real and nominal variables.

Given the modelling approach, what we recover are business cycle regularities rather than structural parameters, and the decomposition that recovers them is not unique: linearity, the assumed component dynamics, orthogonality and our priors together select one approximation to the data, with regime shifts, outliers and non-linearities absorbed by the idiosyncratic components as wedges between data and model. The model should therefore be read not literally but as a device for isolating dominant regularities, and our robustness analysis assesses how well it does so. What survives that scrutiny is a modest claim, but a sharp one: inflation and real activity are not uncorrelated at business cycle frequencies, and the Phillips curve was never 'dead'.

Related literature. Our modelling approach follows the tradition of semi-structural models that combine reduced-form statistical methods with theory-informed restrictions to identify unobserved quantities of interest (for a review, see Hasenzagl et al., 2025). Examples include work in the trend-cycle tradition of Harvey (1985), but also VARs (Del Negro et al., 2017) and factor models (Barigozzi and Luciani, 2023) with stochastic trends.[3]

In the context of this literature, our trend-cycle model imposes a minimal set of multivariate restrictions, which lets us single out cleanly the business cycle commonalities that are the object of this study. In this sense the exercise identifies stylised facts in the tradition of the filtering literature (Stock and Watson, 1999; Canova, 1998), and is closer in spirit to a well-calibrated band-pass filter (Baxter and King, 1999) than to a fully identified VAR or an estimated DSGE model.

Within the trend-cycle tradition our closest antecedent is Hasenzagl et al. (2022), from which we depart in three ways. First, our specification is at once more flexible – potential output has a stochastic growth rate, modelled as a random walk, so that trend growth can itself change over time – and leaner, in that it features a single common cycle, the business cycle.[4] Second, we bring the short-term policy rate into the multivariate system, so that the common cycle is disciplined by, and transmits to, the systematic component of monetary policy. Third, we take a long view: a century of US data lets us ask how stable business cycle regularities are across

[3]Relevant references include Morley et al. (2003) and Grant and Chan (2017) (US output gap), Mertens and Nason (2015) and Mertens (2016) (US trend inflation and inflation dynamics), Jarociński and Lenza (2018) (euro-area output gap), Hasenzagl et al. (2022) (US trend inflation, output gap and the Phillips curve), Ascari and Fosso (2024) and Bianchi et al. (2026) (US Phillips curve), Maffei-Faccioli (2020) (US potential output), and Zaman (2021) (US long-run equilibrium levels).

[4]In Hasenzagl et al. (2022) potential output is a random walk with constant drift, and the model features two common cycles: the business cycle and an energy price cycle. We dispense with the latter.

monetary regimes, institutional change and structural breaks, a question that postwar samples cannot address.

Two papers in the VAR literature reach conclusions that contrast with ours. Using a max-share identification of the shock that accounts for the bulk of the business cycle variance of US real activity, Angeletos et al. (2020) find that it drives most real fluctuations but contributes little to inflation, or to TFP, at any frequency. Del Negro et al. (2020) show that this disconnect emerges only around 1990, when inflation stops responding to the shock while unemployment responds as before, and, through an estimated New Keynesian DSGE model, attribute it mainly to a flatter structural Phillips curve, with monetary policy playing a secondary role. We share the view that a common component drives most business cycle comovement, but rather than identifying a shock, or a bundle of shocks, we extract a common dynamic component that no single shock is required to explain. This matters for subsample comparisons, which, as Sims (2020) observes, can mistake changes in the bundle of shocks for changes in the correlations among variables, a variation our idiosyncratic components absorb by construction. The treatment of trend inflation also accounts for much of the difference: holding it constant, our reduced-form Phillips curve turns positive over 1946–84 and becomes nearly vertical over 1985–2019.

Closest to our findings is Bianchi et al. (2026), who note that a fixed-coefficient VAR may fail to separate business cycle from low-frequency movements over short samples with structural breaks. Within a trend-cycle VAR, they find that the same max-share shock accounts for a substantial share of the inflation cycle, and the Phillips curve is no longer flat. Our results point the same way: once low-frequency movements are removed, cyclical inflation comoves with cyclical activity. The exercises differ in object and scope. They condition on a single identified shock in postwar data; we extract a common component of seven series, including expectations and the policy rate, identified by their comovement rather than by the variance of any one of them, and use a century of data to study the stability of the reduced-form Phillips curve across monetary regimes, institutional change and structural breaks.

Low-frequency movements also connect our findings to Hazell et al. (2022), who find a small structural slope in a panel of US states, already small in the early 1980s and only modestly lower since. Their design and ours exploit opposite sources of variation. By including time fixed effects, they remove by construction all aggregate business cycle variation, long-run inflation expectations, monetary and fiscal policy, and aggregate demand and supply shocks. What they estimate is therefore a regional Phillips curve, in an economy where labour, goods and capital move across states. Their aggregate slope does not come from the data but from aggregation within a structural model, under the assumption that regional deviations share the dynamics of the national aggregates. We exploit precisely the aggregate variation their design discards, and recover the reduced-form comovement of inflation and slack along the business cycle. The two estimates answer different questions; we nonetheless share their conclusion that controlling for low-frequency movements in inflation expectations is essential to recover the cyclical relation.

On the policy rate, our results speak to recent work on whether central banks respond differently to demand- and supply-driven inflation. Hofmann et al. (2024) estimate that, since

Volcker, the Federal Reserve has responded about four times more strongly to the former than to the latter, and Nakamura et al. (2025) document sizeable departures from the Taylor rule, including the Federal Reserve's decision to look through part of the post-pandemic surge. Our evidence is consistent with these findings: the cyclical component of the policy rate is largely accounted for by the common, demand-like cycle, while its response to the idiosyncratic component of inflation, which may capture supply-driven movements, is small and short-lived (Section 4). At the same time, our results point to a relatively stable systematic component of monetary policy over long spans of time, across what were potentially different policy frameworks.

Finally, our model is linear with constant coefficients, whereas a growing literature points to state dependence in price setting and non-linearities in the Phillips curve. Cavallo et al. (2024) show that large cost shocks raise the frequency of price changes and hence their pass-through, and Gitti (2025) finds that the regional Phillips curve is flat in slack labour markets and steepens markedly in tight ones. In our framework such non-linearities are absorbed by the idiosyncratic components, and our slopes are averages over the states visited in each sample. Their similarity across subsamples that include high-inflation episodes suggests that state dependence has not systematically shifted the reduced-form relation at business cycle frequency, although our approach is not designed to detect it.

2 A stylised NK model with trends and cycles

This section develops a stylised model to organise the empirical analysis. It shows that a standard New Keynesian description of the macroeconomy, augmented with the stochastic trends that generate its low-frequency movements, admits under rather standard assumptions a reduced-form trend-cycle representation. In that representation the economy is summarised by a small number of stochastic trends and by a single cyclical state which transmits to prices through a reduced-form Phillips curve and to the labour market through Okun's law, and on which every other observable loads. We do not claim that this representation describes the economy exactly. We take it as an approximation to the dynamic relationships that hold at business cycle frequencies.[5]

2.1 A stylised New Keynesian model

Trends and cycles. At the very core of the study of business cycle fluctuations there is a decomposition of output (in log levels) into a trend – potential output – and a stationary component:

$$y_t = \tau_t^y + \widehat{y}_t. \tag{1}$$

The trend, τ_t^y, is usually thought of as determined by technological progress and by demographic and institutional factors, which inform the long-run behaviour of GDP. A commonly adopted specification for potential output is an integrated process of order two, i.e. I(2), in which its

[5]Details of the derivations are reported in full in Online Appendix A.

growth rate $g_t \equiv \Delta\tau_t^y$ is a driftless random walk that captures stochastic changes in trend growth:

$$\tau_t^y = \tau_{t-1}^y + g_t, \tag{2}$$

$$g_t = g_{t-1} + u_t^g. \tag{3}$$

The fluctuation of output off its trend, $\widehat{y}_t$, is instead due to the action of different structural cyclical factors – demand, supply, monetary – pushing output off its long-run equilibrium. This measure of business cycle fluctuations, in the Frisch–Slutsky paradigm, is usually considered to be representable as a mean-zero stationary stochastic process.

Similarly, we decompose unemployment into a trend and a cyclical component,

$$u_t = \tau_t^u + \widehat{u}_t. \tag{4}$$

Trend unemployment, τ_t^u – the rate consistent with output at its potential, commonly referred to as equilibrium unemployment or NAIRU (Non-Accelerating Inflation Rate of Unemployment) – is thought to be due to structural and institutional factors in the labour market. We model it as a driftless random walk,

$$\tau_t^u = \tau_{t-1}^u + u_t^{\tau,u}. \tag{5}$$

We do not explicitly model the labour market, but we assume that the cyclical component of output is reflected in the cyclical component of unemployment via Okun's law, i.e.

$$\widehat{u}_t = -\delta_u(L)\widehat{y}_t, \tag{6}$$

where the lag polynomial $\delta_u(L)$ has non-negative coefficients, so that a positive deviation of output from its trend is associated with unemployment below the NAIRU.

Inflation is also characterised by a trend, i.e. the inflation rate prevailing in the absence of cyclical factors, and by a mean-zero cyclical component,

$$\pi_t = \tau_t^\pi + \widehat{\pi}_t. \tag{7}$$

We model trend inflation as a driftless random walk,

$$\tau_t^\pi = \tau_{t-1}^\pi + u_t^{\tau,\pi}, \tag{8}$$

so that its innovations are unpredictable and $E_t\tau_{t+1}^\pi = \tau_t^\pi$. Expected inflation then splits into the current trend and the expected cyclical component,

$$E_t\pi_{t+1} = \tau_t^\pi + E_t\widehat{\pi}_{t+1}, \tag{9}$$

and, iterating, trend inflation reflects the long-term rational expectations of agents, $\tau_t^\pi = \lim_{h\to\infty} E_t\pi_{t+h}$ (Kozicki and Tinsley, 2001; Ireland, 2007; Cogley and Sbordone, 2008; Cogley

et al., 2010). It coincides with the inflation target only when the central bank is fully credible, and the two can differ substantially when credibility is incomplete.

Finally, the nominal interest rate, set by the central bank, is characterised by a nominal trend, which can be decomposed into a real interest rate trend plus the inflation trend, and by a cyclical component,

$$i_t = \tau_t^i + \widehat{\imath}_t = \tau_t^r + \tau_t^\pi + \widehat{\imath}_t. \tag{10}$$

The processes for τ_t^r and for the cyclical component are discussed below.

To summarise our stylised description of the economy, each variable is characterised by stochastic trends and stationary cyclical components – which we assume to be uncorrelated and with unpredictable innovations. While potential output is I(2) and its growth rate g_t is a driftless random walk, trend unemployment and trend inflation are I(1) and assumed to be driftless random walks. These assumptions also imply that the neutral real rate τ_t^r inherits a stochastic trend in the form of a driftless random walk.

The three-equation model. With Calvo pricing, and assuming full indexation to trend inflation of the prices that are not reset, the dynamics of the economy are described by three equations in which the trends cancel out and only cyclical components appear (see Cogley and Sbordone, 2008):

$$\widehat{\pi}_t = \beta E_t \widehat{\pi}_{t+1} + \kappa\, \widehat{y}_t + u_t^s, \tag{11}$$

$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma\big(\widehat{\imath}_t - E_t \widehat{\pi}_{t+1} - u_t^d\big), \tag{12}$$

$$\widehat{\imath}_t = E_t \widehat{\pi}_{t+1} + \theta_\pi \widehat{\pi}_t + \theta_d u_t^d + u_t^{mp}, \tag{13}$$

where u_t^d, u_t^s, and u_t^{mp} are demand, supply, and monetary-policy disturbances, taken to be i.i.d. throughout this section.

The structural Phillips curve, in Eq. (11), is the aggregate supply relationship that relates output fluctuations positively to inflation. The slope of that relationship is given by the parameter κ. In New Keynesian models it is microfounded from the firms' optimal pricing problem and links marginal costs to inflation.[6]

The hybrid IS curve, in Eq. (12), has weights on past and expected future activity that sum to one, with $\alpha \in (0,1)$ the degree of backward-lookingness. It creates a link between real interest rates and real activity, the strength of which depends on the parameter σ.

The policy rule, in Eq. (13), closes the model and captures the behaviour of the central bank. We assume that the policymaker responds to the inflation gap directly. Because the rule already carries expected inflation with a unit coefficient, the Taylor principle takes the form $\theta_\pi > 0$ rather than the more familiar $\theta_\pi > 1$. This stylised description can be seen as a generalisation

[6] We start from the standard New Keynesian Phillips curve written in terms of the output gap, the difference between output and the level that would prevail under flexible prices, in which the disturbance is an exogenous cost-push shock ε_t. Splitting the gap into $\widehat{y}_t$ and $\widehat{y}_t^{flex}$, and noting that $\widehat{y}_t^{flex} = (1+\eta)(\chi+\eta)^{-1}\,\widehat{a}_t$, with η the inverse Frisch elasticity, χ the coefficient of relative risk aversion and $\widehat{a}_t$ the transitory component of technology, delivers Eq. (11) with $u_t^s = \varepsilon_t - \kappa(1+\eta)(\chi+\eta)^{-1}\,\widehat{a}_t$.

of the models proposed by Del Negro et al. (2020) and McLeay and Tenreyro (2020). The rule departs from a standard Taylor rule in two respects: it has no explicit response to the output gap, and it lets the central bank lean directly against demand disturbances, through θ_d, while responding to supply disturbances only through their effect on inflation. It therefore allows for a differential response to demand and supply shocks, in the spirit of the targeted Taylor rule of Hofmann et al. (2024).[7]

It is worth observing that the neutral rate in the model, the real rate consistent with output at potential, turns out to be

$$r_t^n = \frac{1}{\sigma}(1-2\alpha)\, g_t + u_t^d = \tau_t^r + u_t^d, \tag{14}$$

so that the neutral rate depends on trend growth: it is the driftless random walk g_t that gives the neutral rate its permanent component, which in turn coincides with the real rate trend.

The equilibrium solution. Substituting (13) into (12) leaves a system in $(\widehat{\pi}_t, \widehat{y}_t)$ whose only endogenous predetermined variable is $\widehat{y}_{t-1}$. We can therefore guess and verify a solution in which the stationary component of output is the single cyclical state,

$$\widehat{y}_t = \gamma\, \widehat{y}_{t-1} + \textstyle\sum_j b_j u_t^j, \qquad \widehat{\pi}_t = \varphi\, \widehat{y}_{t-1} + \textstyle\sum_j c_j u_t^j, \tag{15}$$

with $j \in \{d, s, mp\}$. Substituting into (11) and (12) and matching coefficients delivers the equilibrium solution for the dynamics of output,[8]

$$\widehat{y}_t = \gamma\, \widehat{y}_{t-1} + \frac{\sigma\gamma}{\alpha}\Big[(1-\theta_d)\, u_t^d - \theta_\pi\, u_t^s - u_t^{mp}\Big], \tag{17}$$

and the reduced-form Phillips curve in the contemporaneous cyclical component of output,

$$\widehat{\pi}_t = \Phi\, \widehat{y}_t + u_t^s, \qquad \Phi \equiv \frac{\kappa}{1-\beta\gamma}. \tag{18}$$

Iterating the equilibrium solutions for output and for inflation, and using the martingale property of the trends, it is also possible to derive an expression for inflation expectations:

$$E_t \pi_{t+h} = \tau_t^\pi + \Phi\gamma^h\, \widehat{y}_t, \qquad h \geq 1. \tag{19}$$

The policy rate inherits the same property: substituting (18) into (13) gives

$$\widehat{\imath}_t = \Phi(\gamma + \theta_\pi)\widehat{y}_t + \theta_\pi u_t^s + \theta_d u_t^d + u_t^{mp}\,. \tag{20}$$

[7] We take this feature to the data in Section 4.4.

[8] The persistence parameter γ is the stable root of the cubic equation

$$F(\gamma) \equiv \beta(1-\alpha)\gamma^3 - (1-\alpha+\beta)\gamma^2 + \big(1+\alpha\beta+\sigma\kappa\theta_\pi\big)\gamma - \alpha = 0, \tag{16}$$

which exists in $(0,1)$ whenever $\theta_\pi > 0$, since $F(0) = -\alpha < 0 < \sigma\kappa\theta_\pi = F(1)$. At $\theta_\pi = 0$ a root sits exactly on the unit circle and the equilibrium fails to be determinate.

A single state thus determines every cyclical component of the system, and Equations (17)–(18) are the model's solution. Proposition A.1 in the Online Appendix A shows that, for $\alpha > 0$, $\kappa > 0$ and $\theta_\pi > 0$, this is indeed the unique bounded rational-expectations equilibrium (Blanchard and Kahn, 1980).

A few observations are in order. First, the strength of the reduced-form curve (18) depends on the structural slope κ and on the conduct of policy. As κ approaches zero the relationship becomes flat, as would arise from structural changes in pricing that sever the link between inflation and marginal costs.

Second, the overall size of the fluctuations in output depends on θ_d, the parameter governing the response of the central bank to demand shocks. For $\theta_d = 1$ the central bank completely offsets demand shocks, and with them the variation that traces out the reduced-form relationship between inflation and the cyclical component of output. Fluctuations in prices would then be due to supply shocks alone, and possibly orthogonal to output: an extreme case of the mechanism in McLeay and Tenreyro (2020), in which the optimal policy response makes the reduced-form Phillips curve disappear in post-Volcker data.

Third, while demand and monetary policy shocks generate a positive correlation between output fluctuations and inflation, supply shocks can generate a negative one. The effects of supply shocks, however, depend on the monetary policy response: absent a response of the interest rate to inflation, they would not affect real variables and would lead only to fluctuations in prices, although they may still affect trend output.

Fourth, demand and monetary disturbances move output and inflation along the curve, whereas supply disturbances move inflation off it and, for $\theta_\pi > 0$, push activity in the opposite direction as the central bank leans against them.[9] Because unemployment and the cyclical component of output move in opposite directions, episodes dominated by supply shocks therefore trace steep – and, for θ_π small, nearly vertical or positively sloped – clouds of unemployment and inflation in the raw data. This is the logic behind Figure 1, and the reason why isolating the cyclical components matters.

A trend-cycle representation. Collecting the trend components and Okun's law in (1)–(10), the solutions (18) and (19), and the policy rate (20), the model maps into an unobserved-components representation in which a single common cycle loads on all variables alongside their trends. We summarise this result in the following lemma.

Lemma 1 (Trend-cycle representation) *Let the economy be described by (11)–(13) with i.i.d. disturbances, let the trends satisfy $E_t\tau^x_{t+1} = \tau^x_t$ for $x \in \{u, \pi\}$ and $E_t g_{t+1} = g_t$, and let $\theta_\pi > 0$, with $\gamma \in (0,1)$; set $\Phi \equiv \kappa/(1-\beta\gamma)$. Then the observables*

$$z_t \equiv \left(y_t,\ u_t,\ \pi_t,\ E_t\pi_{t+h},\ i_t\right)' \tag{21}$$

[9] Setting $u^s_t = 0$ in (18), demand and monetary shocks leave the economy exactly on the line $\widehat{\pi}_t = \Phi\widehat{y}_t$: they trace the curve out. A supply shock instead displaces inflation vertically off it, by u^s_t, and moves the gap in the opposite direction.

admit the representation

$$z_t = \Lambda_\tau \tau_t + \lambda(L)\widehat{y}_t + w_t, \tag{22}$$

where $\widehat{y}_t$*, the single state entering every row, follows the autoregression* (17)*, and where*

$$\Lambda_\tau \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \qquad \tau_t \equiv \begin{pmatrix} \tau_t^y \\ \tau_t^u \\ \tau_t^\pi \\ \tau_t^r \end{pmatrix} \qquad \lambda(L) \equiv \begin{pmatrix} 1 \\ -\delta_u(L) \\ \Phi \\ \gamma^h \Phi \\ \Phi(\gamma + \theta_\pi) \end{pmatrix} \qquad w_t \equiv \begin{pmatrix} 0 \\ 0 \\ u_t^s \\ 0 \\ \theta_\pi u_t^s + \theta_d u_t^d + u_t^{mp} \end{pmatrix}.$$

The proof of this Lemma is reported in full in Online Appendix A. Two features of (22) are what the empirical model of Section 3 inherits. First, the trend loadings Λ_τ are either one or zero, a consequence of the unit-root properties of the trends. Second, a single stationary component enters every observable, so that activity, prices, expectations and the policy rate are all informative about the same latent state; this is the sense in which it is estimated jointly rather than extracted from output alone. This formalises the intuition of Burns and Mitchell (1946) of the business cycle as commonality in the fluctuations of macroeconomic variables.

3 An empirical model of trends and cycles

The stylised model of the previous section motivates an unobserved-components specification that inherits the structure of Eq. (22): variable-specific stochastic trends and a single common business cycle component. Its flexible measurement-equation counterpart is

$$z_t = \widetilde{\Lambda}_\tau \tau_t + \widetilde{\lambda}(L)\, \widehat{y}_t + \widetilde{w}_t, \tag{23}$$

where a tilde denotes an object that is estimated from data. The trend loadings $\widetilde{\Lambda}_\tau$ retain the zero-one pattern of Λ_τ; the lag polynomials in $\widetilde{\lambda}(L)$ connect the cyclical component of each variable to the common business cycle component; and $\widetilde{w}_t$ collects the wedges between the observed data and these relationships. We discuss each component in turn.

The vector of observables, z_t, comprises seven series: real GDP (y_t), the unemployment rate (u_t), headline (π_t) and core (π_t^c) inflation, the one-year-ahead inflation expectations of the SPF ($F_t^{spf}\pi_{t+4}$) and of the University of Michigan survey ($F_t^{uom}\pi_{t+4}$), and the short-term interest rate (i_t).

The vector of trends, τ_t, is specified as a set of independent stochastic trends. Output enters the measurement equation in levels, with its potential τ_t^y modelled as a stochastic trend of order two, i.e. I(2), with a stochastic growth rate g_t^y:

$$\tau_t^y = \tau_{t-1}^y + g_t^y, \qquad g_t^y = g_{t-1}^y + u_t^g. \tag{24}$$

Trend growth is thus a driftless random walk, which is estimated as a by-product.

All the other trends are driftless stochastic trends of order one, i.e. I(1):

$$\tau_t^x = \tau_{t-1}^x + u_t^{\tau,x}. \tag{25}$$

In particular, the unemployment rate has its own stochastic trend τ_t^u, which can be interpreted as the NAIRU. A single trend inflation τ_t^π is estimated as the common low-frequency component shared by headline and core inflation, the two survey expectations, and the nominal interest rate. It is, by construction, the model-based long-run forecast of inflation. The nominal rate is also driven by an independent unit-root component τ_t^r, the equilibrium real rate, which we treat as orthogonal to the other trends. Finally, survey-based inflation expectations can deviate persistently from a rational forecast because of a time-varying bias, which we model as two further driftless random-walk components, $\mu_t^{spf,\pi}$ for professional forecasters' and $\mu_t^{uom,\pi}$ for consumers' expectations. Each loads only on the corresponding survey, so that neither affects the common price trend τ_t^π.

The common business cycle component is modelled as an ARMA(2,1) process, the simplest specification that displays pseudo-cyclical behaviour (Harvey, 1985). It can be written in a VAR(1) representation as

$$\begin{aligned} \widehat{y}_t &= \rho\cos(\lambda)\,\widehat{y}_{t-1} + \rho\sin(\lambda)\,\widehat{y}^*_{t-1} + v_t, \\ \widehat{y}^*_t &= -\rho\sin(\lambda)\,\widehat{y}_{t-1} + \rho\cos(\lambda)\,\widehat{y}^*_{t-1} + v^*_t, \end{aligned} \tag{26}$$

where v_t and v_t^* are uncorrelated white-noise disturbances, $0 \le \lambda \le \pi$ is the cycle frequency, $0 \le \rho \le 1$ is the damping factor (the process is stationary for $\rho < 1$), and $\widehat{y}^*_t$ is an auxiliary state that allows for the VAR(1) representation. Equivalently, the ARMA(2,1) representation is

$$(1 - 2\rho\cos(\lambda)L + \rho^2 L^2)\,\widehat{y}_t = (1 - \rho\cos(\lambda)L)v_t + \rho\sin(\lambda)L\,v_t^*. \tag{27}$$

This single component enters every observable through the lag polynomials $\sum_j \gamma_{i,j}L^j$ that make up the loading vector in Eq. (28). Cyclical fluctuations propagate across indicators both contemporaneously and with lags, allowing the model to match the timing of the comovement as well as its size. The loading on output is normalised to one, which fixes the scale of the common business cycle component, so that it is measured in percentage points of potential output; every other loading is estimated freely. Since the polynomials are unrestricted in sign, nothing in the specification imposes the direction of the comovement between real and nominal variables.

In addition, each observable carries an independent stationary component $\psi_{i,t}$ that absorbs idiosyncratic and structural shocks, misspecification, non-classical measurement error, and other sources of noise. Each is modelled, like the common business cycle component, as an ARMA(2,1) process with its own frequency and damping parameter. These components can be read as empirical wedges between the observed data and the structural relationships.

Given these assumptions, the measurement equation (23) is specified as

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ \pi_t^c \\ F_t^{spf}\pi_{t+4} \\ F_t^{uom}\pi_{t+4} \\ i_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tau_t^y \\ \tau_t^u \\ \tau_t^\pi \\ \mu_t^{spf,\pi} \\ \mu_t^{uom,\pi} \\ \tau_t^r \end{pmatrix}}_{\text{Trends \& biases}} + \underbrace{\begin{pmatrix} 1 \\ \sum_{j=0}^{1}\gamma_{2,j}L^j \\ \sum_{j=0}^{1}\gamma_{3,j}L^j \\ \sum_{j=0}^{1}\gamma_{4,j}L^j \\ \sum_{j=0}^{2}\gamma_{5,j}L^j \\ \sum_{j=0}^{2}\gamma_{6,j}L^j \\ \sum_{j=0}^{2}\gamma_{7,j}L^j \end{pmatrix} \widehat{y}_t}_{\text{Common cycle}} + \underbrace{\begin{pmatrix} \psi_{1,t} \\ \psi_{2,t} \\ \psi_{3,t} \\ \psi_{4,t} \\ \psi_{5,t} \\ \psi_{6,t} \\ \psi_{7,t} \end{pmatrix}}_{\text{Idiosyncratic cycles}} . \quad (28)$$

3.1 A discussion of the model

The empirical model connects closely to the model of the previous section, yet departs from it in important ways. A few remarks are in order. First, the theoretical model implies a number of parametric restrictions on the reduced-form loadings of the cyclical component, which we do not impose. Imposing them would require strong assumptions and would limit the ability of the data to inform the model.

Second, as in the theoretical model, a single stationary process drives business cycle co-movement across all observables. We refer to it as the common business cycle component, the empirical counterpart of the stationary component of output of Section 2. Here, however, it is specified as an ARMA(2,1) so as to display stochastic cyclical behaviour.[10]

Third, the trend components are all independent, a convenient assumption that simplifies estimation. The model of Section 2 features a nominal neutral rate that is a function of trend growth and trend inflation; in the empirical model we do not impose restrictions connecting these trends.

Fourth, in the representation (22) the vector w_t has a non-diagonal variance matrix and is correlated with the common business cycle component. The baseline empirical model instead treats $\widetilde{w}_t$ as a vector of idiosyncratic components with independent innovations, orthogonal to it, again for estimation convenience.

3.2 Bayesian estimation

The model can be cast in linear state-space form and estimated with Bayesian techniques, using an Adaptive Metropolis-within-Gibbs algorithm. We employ the simulation smoother of Durbin and Koopman (2002), with the modification of Jarociński (2015), to condition the estimates of cycles and trends on the full sample. Following Hasenzagl et al. (2022), each series is normalised

[10] The ARMA(2,1) is also the order taken by its theoretical counterpart, the stationary component of output of Section 2, once a disturbance is allowed to be persistent. If the demand shock follows an AR(1) with persistence ϱ, applying the filter $(1-\varrho L)$ to the solution (17) leaves a process whose autocovariances vanish beyond the first lag, so that $(1-\varrho L)(1-\gamma L)\widehat{y}_t = (1+\vartheta L)e_t$: $\widehat{y}_t$ is then exactly an ARMA(2,1), with one autoregressive root inherited from its own endogenous persistence and the other from the persistence of the shock. These two roots are real, rather than the complex-conjugate pair implied by the specification above and by Harvey (1985). See Online Appendix A for the derivations.

Table 1: Prior distributions

Name	Support	Density	Parameter 1	Parameter 2
γ (loadings)	$\mathbb{R}$	Normal	0	1000
τ_0 (initial trend levels)	$\mathbb{R}$	Normal	y_0	1000
σ^2 (shock variances)	$(0, \infty)$	Inverse-Gamma	3	1
σ_g^2 (trend-growth shock)	$(0, \infty)$	Inverse-Gamma	3	0.005
ρ (damping)	[0.001, 0.97]	Uniform	0.001	0.97
λ (frequency)	[0.001, π]	Uniform	0.001	π

Notes: Prior distributions for the model parameters. For the Normal, Parameters 1 and 2 are the mean and standard deviation; for the Inverse-Gamma, the shape and scale; for the Uniform, the lower and upper bounds. The loadings γ take a diffuse Normal. The initial trend levels are centred at the first observation of each series, with a diffuse Normal. Shock and measurement-error variances take an Inverse-Gamma$(3, 1)$, with the single exception of the innovation to trend GDP growth g_t^y, which takes a tighter Inverse-Gamma$(3, 0.005)$. The bounds on the Uniform priors ensure that the stochastic cycles are stationary and correctly specified, following Harvey (1990).

by the standard deviation of its first differences, which places the data on a comparable scale and improves the mixing of the sampler. Missing observations – pervasive in the pre-war sample – are handled through the Kalman filter (Shumway and Stoffer, 1982), which reconstructs the series from the information available at each date. The prior distributions are reported in Table 1, and Online Appendix B reports the full derivation of the sampler.

The loadings γ and the initial trend levels take diffuse Normals, and the cycle damping ρ and frequency λ are uniform over the stationary region (Harvey, 1990), so the persistence and periodicity of the common cycle are left to the data. The shock variances of both the stationary (cyclical) and the non-stationary (trend) components take a weakly informative Inverse-Gamma$(3, 1)$ prior, with the single exception of the innovation to trend output growth, which takes a tighter Inverse-Gamma$(3, 0.005)$.

To see what these prior variances imply for the trends, we follow Bianchi et al. (2026) and report the implied standard deviation of the expected change in each trend over different horizons. At the prior mean, it is about 5.3% over forty years for the annualised trend of GDP growth, about 6.6% over twenty years for the NAIRU, and between roughly 6% and 8% over twenty years for the trend real rate, trend inflation, and the two survey biases. These priors are considerably looser than those of Bianchi et al. (2026), who set the same standard deviations, over the same horizons, to 1%. Even our tightest prior, that on trend output growth, remains looser than the conventional Hodrick–Prescott filter (Hodrick and Prescott, 1997).

In the stability analysis of Section 5.1, the 1985–2019 subsample uses the priors of Table 1, while the 1946–84 subsample requires some regularisation. The two surveys, crucial to pin down trend inflation over the full sample, are sparse before 1984 (the Livingston survey is semi-annual, the Michigan survey starts in 1978), so we exclude them. Until core inflation becomes available in 1957, trend inflation therefore rests on headline CPI alone, whose cycle is distorted by the end of wartime price controls in 1946 and by the Korean War controls of 1951–53. We

therefore give the cyclical components a training-sample prior from a GDP, unemployment and CPI version of the model estimated over 1900–45; the trends and the parameters of core inflation and the policy rate keep the priors of Table 1. Online Appendix C.1 gives the details.

3.3 Extensions & robustness

We consider two extensions of the baseline specification, each relaxing one of its orthogonality assumptions. The first relaxes the orthogonality between the idiosyncratic components of inflation and of the short-term interest rate.

A standard Taylor rule makes the policy rate respond to both the output gap and inflation. In the model of Section 2, the output gap is the stationary component of output, our common business cycle component, net of a transitory technology term that enters the supply shock, while cyclical inflation loads on the common cycle through a lag polynomial, and features its own idiosyncratic term. A standard Taylor rule therefore implies that the policy rate responds to the common cycle and its lags, with coefficients that combine the responses to the output gap and to the common component of inflation, as in our baseline empirical specification. In our estimates the common cycle moves prices and the output gap in the same direction, and is therefore akin to a demand cycle to which the policy rate responds. However, a Taylor rule would also make the policy rate respond to disturbances that move prices and the output gap in opposite directions, such as supply shocks. In our specification such disturbances are absorbed by the idiosyncratic component of inflation. To assess whether the policy rate responds systematically to them, we consider a variant of the model in which the interest-rate equation also loads on the idiosyncratic cycle of inflation, $\psi_{3,t}$, with coefficient θ_π. This specification allows the policy rate to respond differently to demand-driven inflation and to inflation movements that do not follow the demand cycle, such as those caused by supply shocks. A similar idea underlies the targeted Taylor rule of Hofmann et al. (2024). We report the results in Section 4.4.

The second extension relaxes the orthogonality between the innovations to the trends and to the common cycle. In the theoretical model and in our baseline specification, the two are assumed to be mutually orthogonal. Permanent disturbances may, however, move the economy at business cycle frequencies, and transitory shocks to the business cycle may have hysteresis effects on the trends; in either case, the innovations to the trends and to the common cycle would be correlated. We allow for such correlation between the common cycle and the three trends most directly tied to it – trend inflation τ_t^π, trend output growth g_t^y, and the NAIRU τ_t^u – while keeping the trend shocks mutually orthogonal, so that only their correlation with the cycle is freed.

Let ε_t^c be the innovation to the common cycle, v_t in Eq. (26), and u_t^π, u_t^g, u_t^u the innovations to the three trends, with standard deviations $\sigma_\pi, \sigma_g, \sigma_u$. We project the cycle shock on the trend shocks, each in units of its own standard deviation, plus an independent part ν_t:

$$\varepsilon_t^c \;=\; \varphi_\pi \frac{u_t^\pi}{\sigma_\pi} + \varphi_g \frac{u_t^g}{\sigma_g} + \varphi_u \frac{u_t^u}{\sigma_u} + \nu_t, \tag{29}$$

where ν_t is uncorrelated with the trend shocks and has variance σ_ν^2. The coefficients $\varphi_\pi, \varphi_g, \varphi_u$ are the only new parameters and vanish in the orthogonal baseline; because ν_t absorbs the remainder, Eq. (29) defines a valid covariance matrix for any values they take. Since the standardised trend shocks are orthonormal, the correlation between the cycle shock and the shock to trend j is

$$\zeta_j \;=\; \frac{\varphi_j}{\sqrt{\sigma_\nu^2 + \varphi_\pi^2 + \varphi_g^2 + \varphi_u^2}}, \qquad j \in \{\pi, g, u\}. \tag{30}$$

The prior on the coefficients is $\varphi_j \sim \mathcal{N}\big(0, (\omega\,\sigma_\nu)^2\big)$ – centred on the orthogonal baseline, with scale governed by the single parameter ω – while the residual variance σ_ν^2 keeps the Inverse-Gamma$(3, 1)$ prior of the other shock variances (Table 1). Scaling the prior by σ_ν is what makes ω act on the correlation directly: written in terms of the standardised ratios φ_j/σ_ν, the correlation is

$$\zeta_j \;=\; \frac{\varphi_j/\sigma_\nu}{\sqrt{1 + (\varphi_\pi/\sigma_\nu)^2 + (\varphi_g/\sigma_\nu)^2 + (\varphi_u/\sigma_\nu)^2}}, \tag{31}$$

and the prior gives each ratio mean zero and standard deviation ω, so that the induced prior on the correlation depends on ω alone. By simulation, $\omega = 0.1$, 0.2, 0.3 imply a prior standard deviation of the cycle–trend correlation of 0.10, 0.18 and 0.25, respectively. We estimate the model for each of these values and report the results in Section 5.2.

3.4 (More of) a century of data

Table 2: Data and transformations

Variable	**Transf.**	**Frequency**	**Period**	**Source**
Real GDP	Log-levels	Q	1900–1946	Gordon (1986)'s NBER Tables
		Q	1947–2019	FRED
Unemployment rate	Levels	A1901, Q1929	1901–2019	Haver Analytics
Inflation	YoY	A1914, Q1921	1914–2019	Haver Analytics
Core inflation	YoY	Q1957	1957–2019	FRED
Consumers' exp. inflation	Levels	Q1978	1978–2019	University of Michigan
SPF exp. inflation	Levels	S	1946–1983	Livingston Survey
		Q	1984–2019	SPF Philadelphia Fed
Short-term interest rate	Levels	Q1954	1954–2019	FRED

Notes: Macroeconomic variables used in the empirical model. In the frequency column, A, S and Q denote annual, semi-annual and quarterly observations; a letter followed by a year indicates the date from which the series becomes available at that frequency, so that 'A1901, Q1929' denotes a series observed annually from 1901 and quarterly from 1929. 'Consumers' exp. inflation' is the University of Michigan 12-month-ahead expected inflation rate; 'SPF exp. inflation' is the one-year-ahead expected inflation rate from the Survey of Professional Forecasters (the Livingston Survey before 1984).

We use the longest spans of quarterly US data available, for a sample starting in 1900 (Table 2). For the post-World War II period most series are readily available from statistical

agencies and the Federal Reserve; for the earlier sample we rely on historical reconstructions and treat unavailable observations as missing. For real GDP before 1947 we use Gordon (1986), the only source of pre-war quarterly data.[11] When data are available only annually – as for unemployment and inflation in the early sample – we include them as annual observations and treat the intervening quarters as missing. For the one-year-ahead SPF expectation we concatenate the semi-annual Livingston Survey from June 1946 with the quarterly Philadelphia Fed SPF from 1984. For the interest rate we use the quarterly federal funds rate from 1954.

4 Trends and cycles in the US economy

How does the model read US business cycle fluctuations since the beginning of the last century?[12] We first discuss the trends and then the cycles. In reading the results, it is important to stress that the long-run components and the gaps are jointly identified by the model, through the cross-equation restrictions – common trends and a common cycle – and through the parametric forms of the trends and cycles, up to the idiosyncratic wedges and the role played by the priors.

We stress four connected results: (i) the model recovers smooth trends in output and inflation that are broadly in line with the CBO estimate of potential output and, over the sample on which it is available, with the ten-year-ahead inflation expectation; (ii) the business cycle estimated by the model matches the NBER peak and trough dates, as well as the output-gap measures of the CBO and the Greenbooks, albeit with differences in size at the end of the sample; (iii) this common cyclical component accounts for a large share of the cyclical variation in all seven series, carries the signature of a demand cycle, and tightly drives the policy rate; (iv) at business cycle frequency a steep Phillips curve emerges, far steeper than the raw correlation suggests, alongside an Okun's law in line with existing estimates. We document the first three findings in this Section, while we offer a detailed discussion of the last one in Section 5.

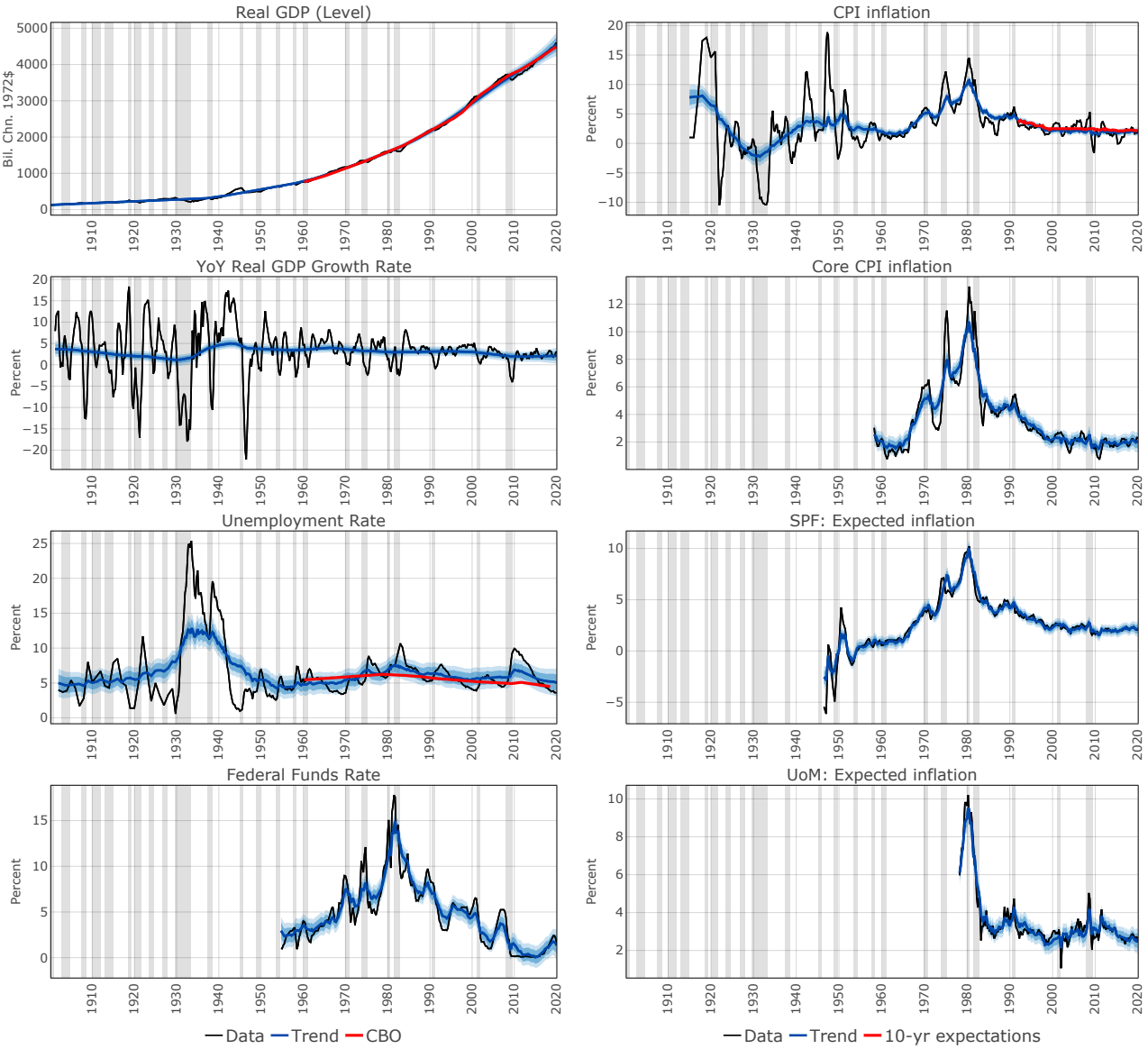
4.1 Long-run components

The model's estimate of potential output (Figure 2a) is smooth and tracks the CBO estimate closely, and its growth rate – the driftless random walk g_t^y – traces the familiar profile of postwar US productivity: fast trend growth through the 1960s, a slowdown from the 1970s, and a further decline in the two decades before the pandemic. The NAIRU drifts slowly, rising through the 1970s and receding thereafter, while the trend component of the short-term rate falls over the last three decades. The CBO estimate of potential output is close to the model's, though higher at the beginning of the common sample and lower at the end of it.

[11] The tables of Gordon (1986)'s *The American Business Cycle*, available from the NBER, are the only existing source of pre-1947 quarterly data, as NIPA quarterly series do not exist before 1947.

[12] The baseline sample ends in 2019. Online Appendix C.5 extends the results to the pandemic period. These estimates are not obtained from a separate estimation: the parameters are held at their 1900–2019 posterior, and the Kalman filter and smoother are run over the extended sample. They therefore show how the trends and cycles identified on a century of pre-pandemic data read the pandemic and the inflation that followed it.

Figure 2: Trend components.



(a) Activity and policy rate

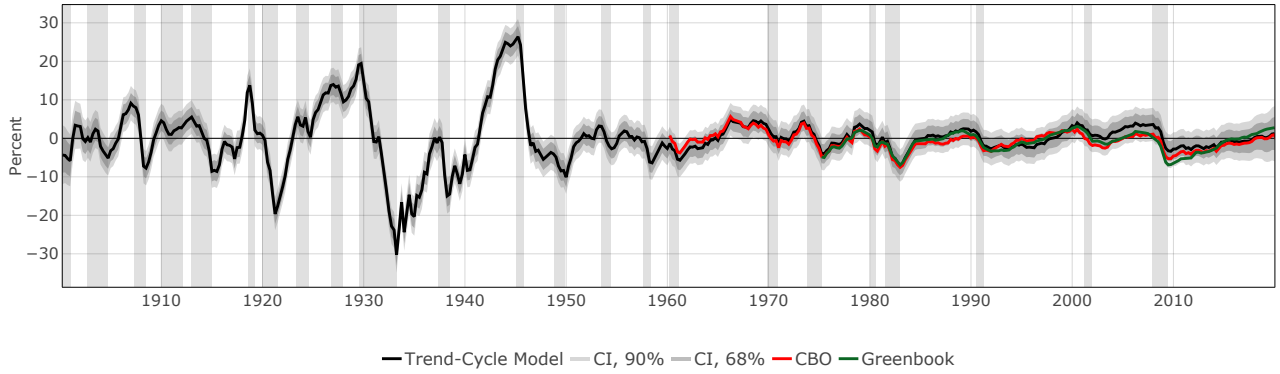
(b) Inflation

Notes: Estimated trends (in blue), with 68% (dark shade) and 90% (light shade) coverage intervals. The observed series are in black, and external comparison measures in red. The model is estimated over the sample 1900–2019.

Trend inflation (Figure 2b) is the common low-frequency component shared by headline and core inflation, the two survey expectations, and the nominal rate. It is stable from the mid-1950s to the mid-1960s, drifts upward through the Great Inflation, and is brought back down by the Volcker disinflation; it then settles at a low and stable level through the 1990s and 2000s. Over the period for which it is available, it closely tracks the ten-year-ahead inflation expectation – an object the model never sees. Since τ_t^{π} is by construction the model-based long-run forecast of inflation, this is the natural external check on the low-frequency block, and the estimate passes it.

4.2 120 years of business cycles

Figure 3: Output gap.



Notes: The model's cyclical component of GDP (common business cycle plus idiosyncratic component, as a percentage of potential output, with 68/90% posterior bands) against the CBO and Federal Reserve (Greenbook) estimates. Shaded areas are NBER recessions.

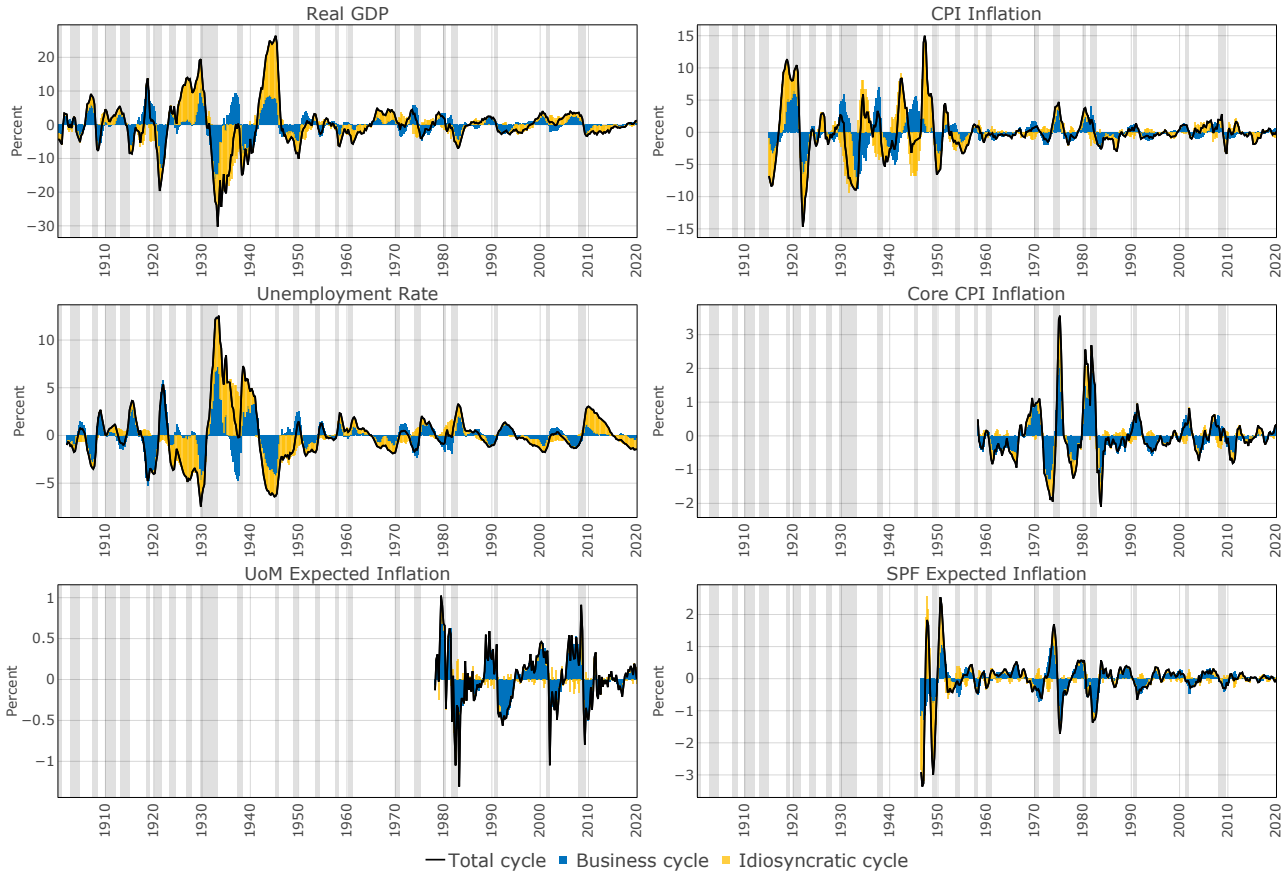
In our empirical model the output gap is the stationary component of output once its trend is removed, which decomposes into the sum of the common business cycle component and the idiosyncratic cyclical component of output, $y_t - \tau_t^y = \widehat{y}_t + \psi_{1,t}$. The common component is the key object: it is the primitive measure of the comovement in business cycle fluctuations, and through its lags it shapes labour-market and price dynamics. Figure 3 plots the model's output gap (black) against the NBER recession dates and compares it with the gap measures of the CBO (red) and the Federal Reserve's Greenbooks (green). The model gap lines up well with the NBER peak and trough dates. Over the span on which all three measures are available, the three gaps are nearly identical until the early 2000s, after which our model signals more overheating before the crisis and less slack after it. The divergence after the 2001 recession reflects a different assessment of potential output: our model's potential absorbs the persistent growth slowdown that begins around the turn of the millennium, so that less of the post-2008 shortfall is read as cyclical (see also Fernald, 2015; Antolin-Diaz et al., 2017).[13]

Across variables, the model's estimated common cycle reproduces the standard business cycle regularities of the US data. In Figures 4 and 5, the blue areas are the common business cycle component, scaled by each series' loading: the cyclical variation that output, unemployment, prices, expectations and the policy rate hold in common, loaded contemporaneously and with lags. The yellow areas are the idiosyncratic, variable-specific components, which capture measurement error, definitional differences, and whatever forms of misspecification the common component cannot absorb.

Along the common cycle, recessions exert downward pressure on prices and expansions push inflation up. The cyclical component of unemployment comoves with the business cycle component, rising as the economy contracts; it lags the common cycle and it is more persistent. In both prices and labour-market variables, the idiosyncratic components are largest in the

[13] Hasenzagl et al. (2022) discuss this divergence between the output gaps produced by models in this class and official estimates, and its implications for the debate on the flattening of the Phillips curve.

Figure 4: Stationary components.



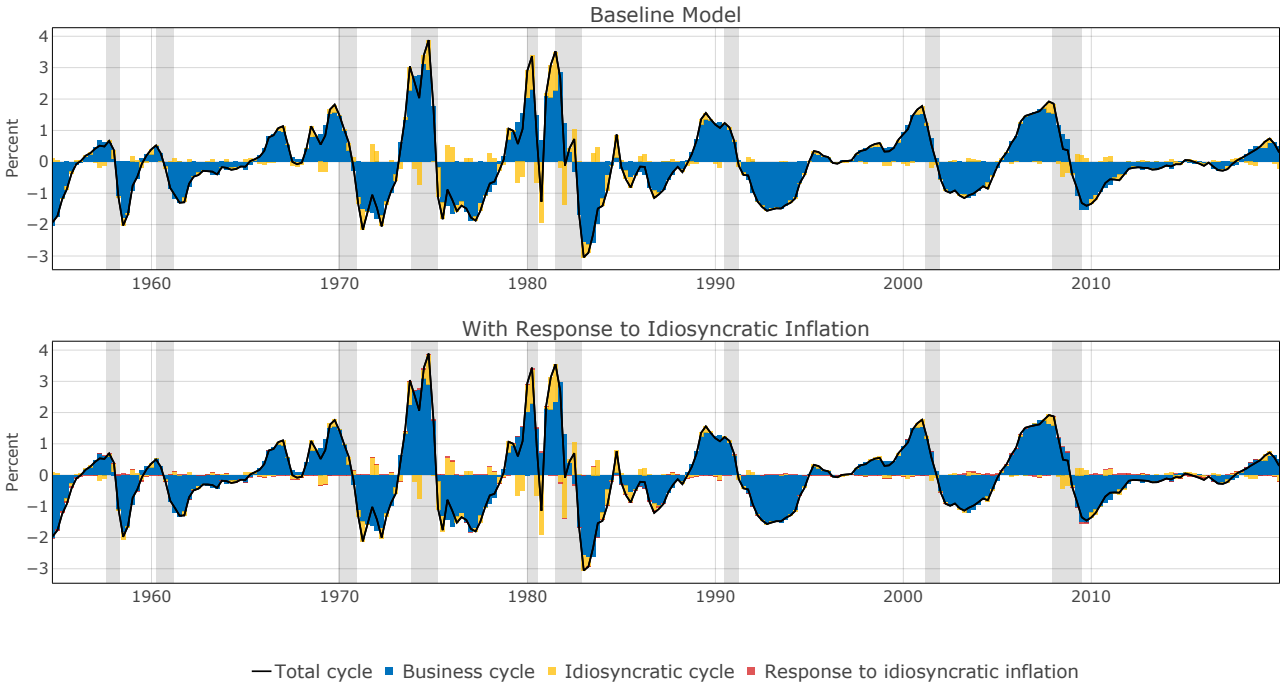
Notes: Cyclical decomposition into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line).

pre-war sample, plausibly reflecting large and unique shocks, structural change, shifts in policy, and the lower quality (or outright absence) of data in those decades.[14]

Let us first examine the pre-World War II period. In this sample, the apparent volatility of the cycle is notably larger than in the postwar period. Before the end of the Second World War, cyclical volatility is driven first by the First World War and the sharp contraction of 1920–21, then by the boom of the late 1920s and the Great Depression, including the relapse of 1937–38, and finally by wartime mobilisation. Unemployment and output share a large common cycle over these decades, but also display extended periods of idiosyncratic dynamics until the end of the 1940s. These plausibly reflect both structural change in the labour market under the exceptional circumstances of the Depression and the war effort, and breaks in measurement. The official series counts workers on emergency relief programmes as unemployed, which raises measured unemployment relative to output in the second half of the 1930s (Darby, 1976); mobilisation then drew millions of workers into the armed forces, outside the civilian labour force.

[14]Part of the difference between the pre- and post-World War II cycles may reflect how the historical series were constructed rather than the economy itself: the extent to which reconstructed pre-war data overstate cyclical volatility has long been debated (Romer, 1986, 1989; Balke and Gordon, 1989). Our quarterly GDP series before 1947 is that of Gordon (1986), which builds on the Balke–Gordon estimates, and unemployment and inflation are observed only annually until 1929 and 1921 respectively.

Figure 5: Cyclical components of the federal funds rate.



Notes: Top chart – baseline model. Bottom chart – model with a systematic policy response to the idiosyncratic inflation cycle. Business cycle in blue, the FFR idiosyncratic cycle in yellow, the response to idiosyncratic inflation in red, and the total cycle as the black line.

Inflation follows a similar pattern: before the end of the Second World War its cyclical component is both more volatile and more idiosyncratic than in the postwar decades. The early 1920s illustrate the point. Inflation reached extreme levels during the First World War and the boom that followed it, peaking around 1920, and the recession of 1920–21 produced a deflation far larger than the contraction in activity can account for (see also Romer, 1988). The model does not force this episode onto the common cycle: it assigns a large part of the swing to the idiosyncratic component of inflation, and part of the preceding run-up to the trend.[15] During the Second World War, by contrast, the idiosyncratic component largely offsets the upward price pressure coming from the common cycle, most likely reflecting wartime price controls (Rockoff, 1984).

To better understand inflation dynamics, we turn to the monetary regime, since our sample spans several distinct ones. Until 1914 the United States was part of the sterling gold standard centred on the London market, formalised domestically by the Gold Standard Act of 1900, without a central bank and with recurrent banking panics; the panic of 1907 led to the founding of the Federal Reserve in 1913. The international system collapsed with the First World War. In the 1920s the Federal Reserve operated within a reconstructed gold-exchange standard, guided largely by the real-bills doctrine (Meltzer, 2003). In this period inflation remained volatile – economic contractions reduced the supply of real bills and hence tightened credit – but around

[15] This attribution should be read with caution. Before 1946 trend inflation is identified from headline inflation alone, with no survey expectations to anchor it and with only annual observations until 1921. Its temporary rise during the war thus reflects the limited information available as much as any shift in long-run expectations under the gold standard.

a lower average than during the war, as the estimated trend shows.

The system collapsed again in the early 1930s: Britain left gold in 1931, and the United States suspended convertibility in 1933 and devalued the dollar in 1934. From 1942 to the Treasury–Fed Accord of 1951, monetary policy was subordinated to the financing of the war debt through a peg on Treasury yields. Inflation volatility rises with the Depression and remains high through the war, decontrol and the Korean War, subsiding only in the early 1950s. This coincides with the Accord, which ended the Federal Reserve's commitment to peg long-term Treasury yields and separated monetary policy from debt management. The model thus reads the high inflation volatility of the first half of the century as the joint product of the business cycle, of idiosyncratic disturbances, and of shifts in trend across peace and war and across monetary regimes.

We now turn to the postwar decades. The output gap (Figure 3) and the common cycle (Figure 4) show a marked decline in the volatility of the US economy and in the size of the idiosyncratic components: through the lens of the model, the variables move in a more common pattern. The lower volatility of the postwar business cycle plausibly reflects structural changes in the economy – a larger government and automatic stabilisers, deposit insurance and the end of banking panics, the economy maturing with a shift from agriculture and manufacturing towards services – as well as smaller shocks (Romer, 1999). Monetary policy is also likely to have played an important role: as the model of Section 2 shows, the systematic response of policy shapes the amplitude of the cyclical responses of the variables to shocks.

The Bretton Woods era, from the late 1940s onward, tied the dollar to gold until President Nixon closed the 'gold window' in 1971, opening a fiat regime.[16] The first postwar years are dominated by the removal of price controls, which the model reads as a large idiosyncratic shock to prices over 1946–48. The Korean War brings a second burst of nominal volatility, concentrated in 1950–51, with a renewed role for idiosyncratic price dynamics and a new round of price controls. The rest of the 1950s is a period of price stability – though not of real stability, with three recessions in the decade – which the historical evidence attributes to an active and systematic monetary response to inflation (Romer and Romer, 2002). Our model captures this stability as cyclical inflation moving closely with the common cycle, with only a small idiosyncratic component. Trend inflation is equally stable until the mid-1960s (Figure 2b).

The following decades saw a resurgence of price volatility. Inflation began rising in the mid-1960s and accelerated after the end of the dollar's link to gold, culminating in the Great Inflation of the 1970s and the Volcker disinflation from 1979 (Meltzer, 2009). Nominal volatility did not, however, return to pre-war levels, nor was it accompanied by a return to pre-war real volatility. From the mid-1980s monetary policy increasingly followed a systematic interest-rate rule aimed at low and stable inflation, a commitment made explicit with the adoption of a 2% inflation target in 2012.

These regime changes match the patterns the model uncovers. From the mid-1960s until

[16] The 'gold window' was the US Treasury's commitment to convert dollars held by foreign central banks and governments into gold at the official price of $35 an ounce.

the Volcker disinflation, trend inflation drifts upward while cyclical inflation largely continues to track the gap, with the exception of the oil shocks of 1973–74 and 1979–80, which the model assigns in good part to the idiosyncratic component. Stability in both returns with the Great Moderation from the mid-1980s, and is tested again after the financial crisis, when cyclical inflation is hit by large swings while the trend remains anchored.

4.3 The US business cycle

We can now pause and take stock of the regularities and historical patterns uncovered by the model. Over the entire sample, the common component (Table 3) accounts for over 40% of the cyclical variation of the three series that cover the pre-war period, and for a higher share of the others, presumably reflecting the higher degree of commonality in the postwar period. Importantly, the common cycle also accounts for a dominant share of the cyclical variation in the short-term interest rate, at 77%.

Table 3: Common business-cycle share of the cyclical decomposition (%).

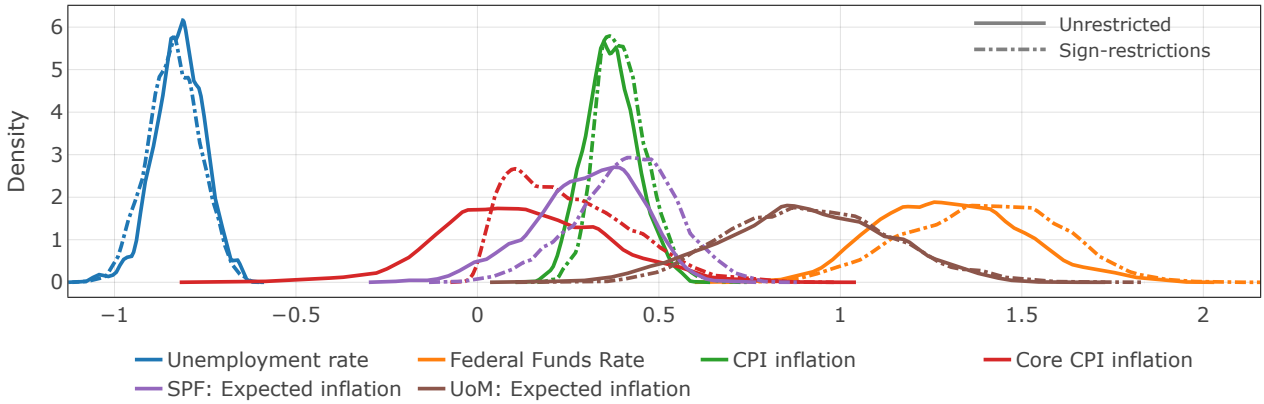
y_t	u_t	π_t	π_t^c	$F_t^{spf}\pi_{t+4}$	$F_t^{uom}\pi_{t+4}$	i_t
41.8	46.1	45.2	61.1	50.5	65.3	77.0
[36.4, 47.7]	[40.3, 52.9]	[39.2, 50.0]	[43.8, 73.2]	[38.0, 62.0]	[50.0, 74.8]	[71.8, 81.0]

Notes: Share of each observable's cyclical decomposition attributable to the common business cycle rather than its idiosyncratic cycle, computed for each posterior draw over every quarter of the series' sample — from its first to its last observation, including the quarters the smoother interpolates over the early annual sample. Reported is the posterior median with the 90% band in brackets.

Along the business cycle, inflation comoves positively with the common business cycle component, while unemployment moves inversely with it, so that inflation and slack move in opposite directions, in relations akin to the classic Phillips curve and Okun's law. Figure 6 reports the posterior distributions of the contemporaneous loadings of each series on the common cycle, which underlie these relations. It is important to stress that the baseline specification of the model leaves the loadings unrestricted, and therefore does not attach an interpretation to the common component. The estimated loadings nonetheless have the pattern of a demand cycle: negative for unemployment, and positive for prices, expectations and the policy rate. Re-estimating the model under sign restrictions that impose this demand pattern on the contemporaneous loadings leaves the posteriors essentially unchanged (Figure 6, dash-dotted lines), with the exception of core inflation, whose loading the data pin down least well.[17]

[17] We restrict the contemporaneous loading on unemployment to be negative and those on headline and core inflation to be positive, leave the loadings on the two survey expectations and the federal funds rate free, and keep the loading on output normalised to one. The restrictions enter as a truncation of the diffuse Normal prior of Table 1, implemented in the Metropolis-within-Gibbs sampler by rejecting draws that violate them; since they are one-sided and apply to one loading at a time, the truncated region is a rectangle and no further identification device is needed. Core inflation has both the shortest sample, beginning in 1957, and by far the smallest cyclical amplitude of the seven series.

Figure 6: Loadings on the common business cycle component.



Notes: Posterior distributions of each series' loading on the common business cycle: the baseline model (solid) against the sign-restricted demand model (dash-dotted).

4.4 Supply and demand in postwar monetary policy

We now zoom in on the postwar conduct of monetary policy. A striking result is that the common business cycle component explains a dominant share of the cyclical movements in the policy rate (Figure 5 and Table 3). Could this be an artefact of the empirical specification? To answer this question, we compare two specifications of the model.

Figure 5 reports the cyclical component of the federal funds rate under both specifications. In the baseline (top panel), the rate loads only on the common cycle. In the alternative specification (bottom panel), it also responds to the idiosyncratic cycle of inflation, which may capture inflation movements that do not follow the pattern of the demand cycle, such as those caused by supply shocks. This specification therefore allows for a differential response to demand and supply shocks (see Section 3.3). The cyclical rate is then

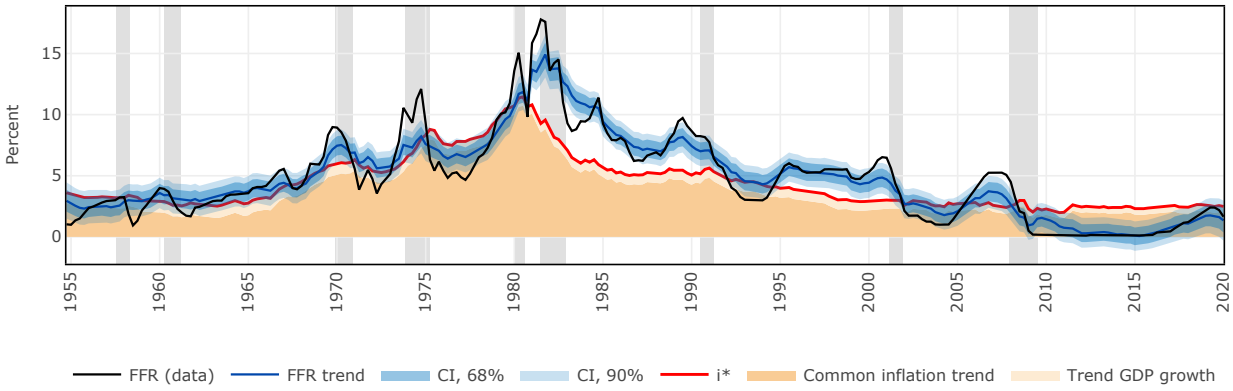
$$i_t - \tau_t^{\pi} - \tau_t^{r} = \Big(\textstyle\sum_{j=0}^{2} \gamma_{7,j} L^j \Big) \widehat{y}_t \;+\; \theta_{\pi}\, \psi_{3,t} \;+\; \psi_{7,t}. \tag{32}$$

The left-hand side is the cyclical component of the policy rate: the observed nominal rate net of its two trends, trend inflation τ_t^{π} and the equilibrium real rate τ_t^{r}. It is therefore the gap between the real policy rate, measured with respect to trend inflation, and its equilibrium level, which is the model's measure of the policy stance. On the right-hand side, the distributed lag $\sum_{j=0}^{2} \gamma_{7,j} L^j$ is the systematic response to the common business cycle component; θ_{π} is the response to the idiosyncratic inflation cycle $\psi_{3,t}$, the part of cyclical inflation that the common component does not account for; and $\psi_{7,t}$ is the stationary idiosyncratic cycle of the interest rate.

In both specifications, the cyclical component of the interest rate is largely accounted for by the common business cycle, with a small idiosyncratic part confined mostly to the Volcker disinflation. The two specifications are nearly indistinguishable: the estimated response to idiosyncratic inflation is small and short-lived, and the total cyclical rate is virtually unchanged. This suggests that the Federal Reserve has responded mainly to the inflation that comes with

the demand cycle and has, to some extent, looked through idiosyncratic movements in inflation, consistent with Hofmann et al. (2024) and Nakamura et al. (2025). In our system, the policy rate therefore moves closely with the common cycle, like the other cyclical indicators, rather than acting as an independent source of fluctuations.

Figure 7: The federal funds rate and the natural nominal rate.



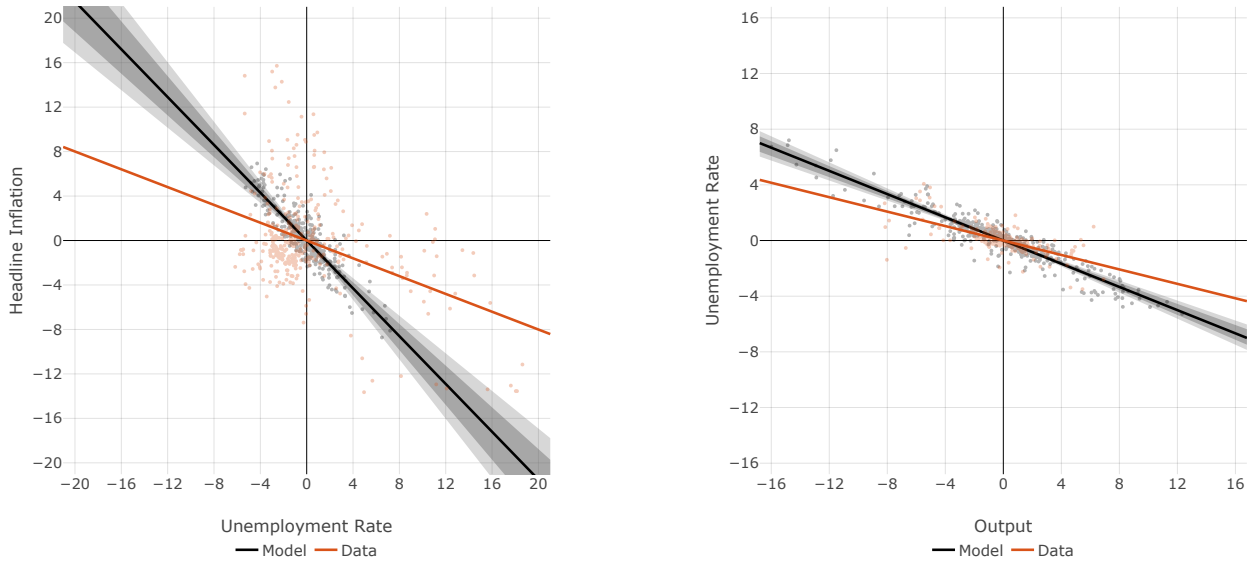
Notes: Federal funds rate (black) and its estimated trend $\tau_t^\pi + \tau_t^r$ (blue, with 68% and 90% posterior bands). The red line is $i_t^* = \tau_t^\pi + g_t$, shown as stacked areas for trend inflation and trend GDP growth (posterior medians).

The cyclical component tells only part of the story of the policy rate; the rest lies in its trend, $\tau_t^\pi + \tau_t^r$. We conclude this section by comparing the trend of the policy rate with a natural nominal rate (Figure 7), defined as $i_t^* \equiv \tau_t^\pi + g_t$: trend inflation plus trend growth, in a Fisher-type approximation of Eq. (14). Several features, in line with the economic history of the United States, stand out. Until the late 1960s the two lines nearly coincide, indicating a policy stance compatible with price stability. In the second half of the 1970s the trend of the policy rate falls about 1 percentage point below i_t^*, at the edge of the 90 percent band, as trend inflation rose faster than the policy rate, consistent with persistently accommodative policy during the Great Inflation. The pattern reverses under Volcker, with persistently tight policy: the trend of the policy rate rises well above i_t^*, by 4 to 5 percentage points over 1981–85. The gap then narrows but persists, at about 2 points in the late 1980s and 1 to 2 points through most of the 1990s, while trend inflation falls from about 10 percent in 1980 to 2 percent in 2000. After 2008 the trend of the policy rate falls about 2 points below i_t^*, and the gap narrows only as policy normalises after 2015, while i_t^* itself declines only slightly. Unlike in the 1970s, however, trend inflation remains anchored at about 2 percent, so the gap does not signal the kind of accommodation that de-anchored expectations during the Great Inflation. Rather, it suggests that after 2008 cyclical factors persistently pushed the economy below its potential, requiring accommodative policy for an extended period. Consistent with this reading, with the policy rate at the effective lower bound from 2008 to 2015, the Federal Reserve turned to unconventional monetary policy to provide further stimulus (for a complementary, structural interpretation, see the literature on the decline of the natural rate, e.g. Holston et al., 2023; Del Negro et al., 2017).

5 Phillips curve and Okun's law

The previous section showed that a single common cycle accounts for a large share of the cyclical variation in activity, unemployment, prices and the policy rate, and that its loadings have the pattern of a demand cycle. We now ask what this component implies for the two classic reduced-form relationships of macroeconomics: the Phillips curve, linking inflation to unemployment, and Okun's law, linking unemployment to output. We read both directly off the model, by regressing the business cycle component of one variable on that of another. Because these components are net of trends and of idiosyncratic wedges, the resulting slopes measure the comovement of the variables at business cycle frequency, free of the low-frequency drift in trend inflation, the NAIRU and potential output, and of variable-specific disturbances such as supply shocks. Since every business cycle component is a distributed lag of the same common cycle, the slopes summarise the relative loadings of the variables on it and the timing of their responses. We first present the estimates over the full century, and then assess their stability across subsamples and rolling windows.

Figure 8: A century of the Phillips curve and Okun's law.



(a) Phillips curve

(b) Okun's law

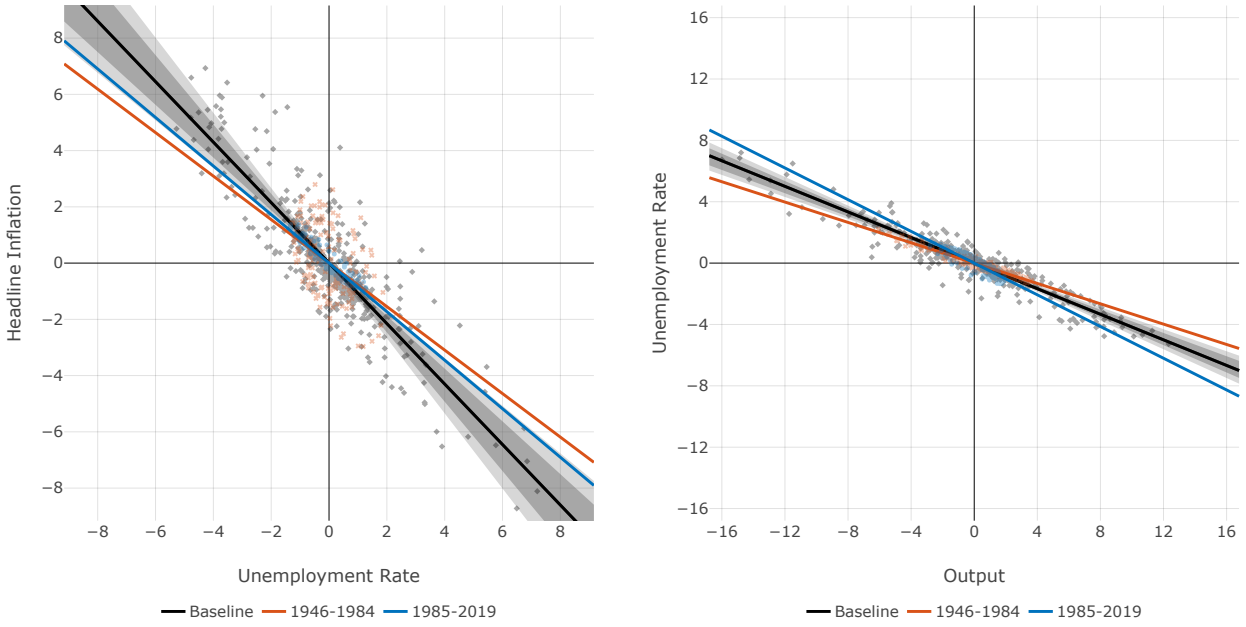
Notes: In black: the business cycle component of one variable against the business cycle component of another, their bivariate linear-regression line, and the 68/90% posterior credible band. In red: the corresponding demeaned raw data and their bivariate linear-regression line.

Figure 8 compares the model-based relationships (black) with the same relationships in the demeaned raw data (red), and delivers the paper's main result. Once low-frequency and idiosyncratic movements are removed, the Phillips curve is far steeper than the raw correlation suggests: the slope of inflation on unemployment is -1.08 over the full century, nearly three times the -0.40 in the raw data. Okun's law steepens too, though by less: the model-based slope is -0.417, essentially the -0.421 of Ball et al. (2017), against -0.259 in the raw data. Combining the two slopes, a one-percentage-point increase in the common business cycle component is associated with a rise of about 0.45 percentage points in cyclical inflation.

5.1 Stability of the Phillips curve and of Okun's law

We assess the stability of the Phillips curve and of Okun's law in two ways: first by fully re-estimating the model on the two postwar subsamples, then on rolling windows, and in each case recomputing the two relationships from the re-estimated business cycle components. Figure 9 reports the reduced-form slopes by subsample, and Table 4 the corresponding 90% credible intervals.[18] The point estimates do move. The Phillips curve is somewhat flatter in both postwar subsamples than over the full century, at −0.77 over 1946–84 and −0.86 over 1985–2019 against −1.08, while Okun's law is at −0.33 and −0.52 against −0.42. The 1946–84 Phillips slope lies just outside the full-sample band, and the 1985–2019 Okun slope clearly outside it. As reported in Table 4, each subsample estimate of the Phillips curve is nonetheless consistent with the full-sample estimate: the subsample bands are wider and all of them contain the corresponding full-sample slope. Strikingly, the Phillips-curve slope is virtually the same before and after 1985: comparing the two postwar subsamples, we find no evidence of the flattening often reported for the Great Moderation.

Figure 9: Phillips curve and Okun's law in different subsamples.



(a) Phillips curve

(b) Okun's law

Notes: In each panel: the full-sample model in black, with its 68/90% posterior credible band in grey, against the two postwar subperiods, 1946–1984 and 1985–2019.

We finally re-estimate the full model on rolling eighty-year windows, from a first window ending in 1980 to a last ending in 2019, and track the reduced-form slopes at business cycle

[18] The output gap and the common cycle estimated on each subsample track the full-sample estimates closely over the periods they share, apart from the first years of the 1946–84 subsample, where the estimates are affected by initial conditions. The posteriors of the cycle's frequency and persistence overlap substantially across samples, with the 1985–2019 posterior placing the cycle at a somewhat lower frequency and higher persistence, consistent with the longer expansions of the period. The innovation variance is markedly lower over 1985–2019. The corresponding charts are Figures C.13 and C.14, reported in Online Appendix C.

Table 4: Reduced-form Phillips-curve and Okun's-law slopes, by specification.

Specification	**Phillips Curve**		**Okun's Law**	
Baseline	-1.08	[-1.34, -0.85]	-0.417	[-0.468, -0.359]
1946–1984	-0.77	[-1.32, -0.28]	-0.332	[-0.380, -0.280]
1985–2019	-0.86	[-1.73, -0.37]	-0.517	[-0.642, -0.358]
Trend Cycle Correlation ($\omega = 0.1$)	-1.08	[-1.36, -0.84]	-0.434	[-0.489, -0.376]
Trend Cycle Correlation ($\omega = 0.2$)	-1.20	[-1.53, -0.93]	-0.447	[-0.515, -0.379]
Trend Cycle Correlation ($\omega = 0.3$)	-1.22	[-1.53, -0.95]	-0.452	[-0.527, -0.384]
Raw data	-0.40	[-0.59, -0.21]	-0.259	[-0.329, -0.189]

Notes: Reduced-form slopes at business-cycle frequency. The Phillips-curve slope is the coefficient of the headline-inflation cycle on the unemployment cycle; the Okun's-law slope is the coefficient of the unemployment cycle on the output gap, both computed from the common business-cycle components. Point estimates are computed on the posterior-median cyclical components, with the 90% posterior interval in brackets alongside. The *Raw data* row reports the corresponding OLS slopes on the demeaned observed series (headline inflation on unemployment; the change in unemployment on GDP growth), with the 90% confidence interval based on HAC (Newey–West) standard errors.

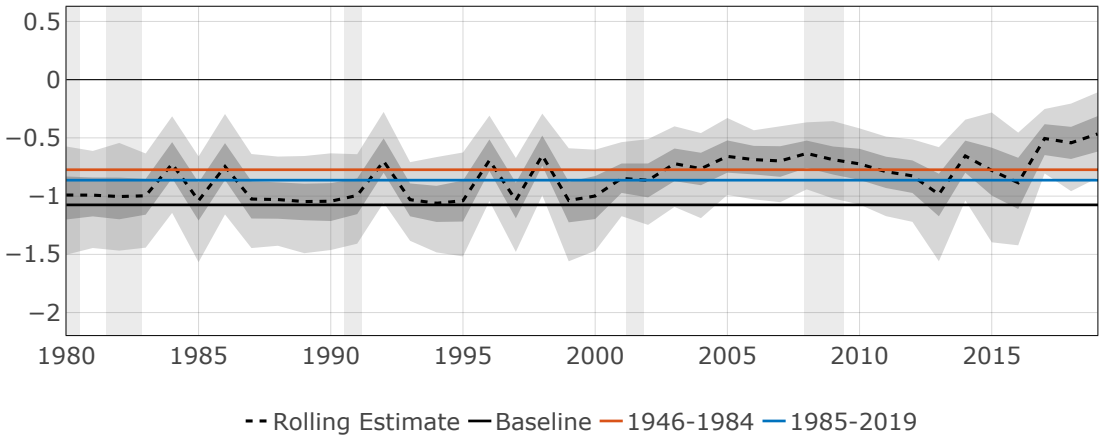
frequency (Figure 10). The two slopes behave differently as the window rolls forward. Okun's law is the more stable of the two: the rolling estimate is somewhat steeper than its full-sample value for windows ending between the mid-1990s and about 2012, and somewhat flatter thereafter, but the full-sample estimate remains inside its 90% band throughout. The rolling Phillips slope becomes flatter from the windows ending in the early 2000s, with point estimates ranging from about -1.1 to -0.5. For most windows ending between about 2002 and 2012, and again from 2016, its 90% band no longer covers the full-sample estimate, although until the final windows it still covers the two postwar subsample estimates. Nonetheless, the slope remains firmly negative: in every window, its 90% band excludes zero.

5.2 Correlated trend–cycle shocks

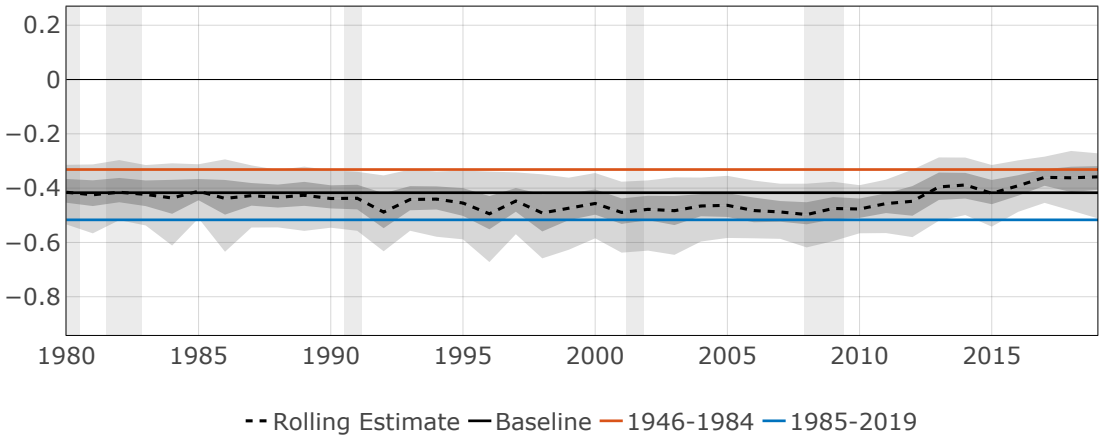
We now turn to the specification of Section 3.3, which relaxes the orthogonality between the innovations to the common business cycle and those to trend inflation τ_t^π, trend output growth g_t^y and the NAIRU τ_t^u. Figure 11 and Online Appendix Figure C.15 compare the reduced-form relationships, the output gap and the common cycle under this specification with the baseline, for each value of the prior scale ω. The conclusions are unaffected: at every prior scale, the Phillips-curve and Okun slopes lie within the baseline posterior band, and the estimated cycle is visually indistinguishable from the baseline.

The slopes do shift, however, systematically and in one direction. Relaxing orthogonality makes both relationships steeper, weakly monotonically in the prior scale: the Phillips slope moves from -1.08 in the baseline to -1.08, -1.20 and -1.22 as ω rises from 0.1 to 0.3, and the Okun slope from -0.417 to -0.434, -0.447 and -0.452 (Table 4). The baseline is therefore, if anything, conservative. The shifts are also small: at $\omega = 0.3$, the Phillips slope moves by 0.14 and the Okun slope by 0.035, about a fifth of the gap between the model-based and the raw

Figure 10: Rolling-window estimation.



(a) Phillips curve



(b) Okun's law

Notes: Reduced-form slopes at business cycle frequency estimated on rolling 80-year windows, plotted against the end year of each window, from 1980 to 2019; the model is fully re-estimated on each window. The dashed black line is the rolling estimate, with 68% (dark grey) and 90% (light grey) posterior bands. The horizontal lines are the slopes estimated over the full sample, 1900–2019 (solid black), and over the subsamples 1946–1984 (orange) and 1985–2019 (blue). Shaded areas are NBER recessions.

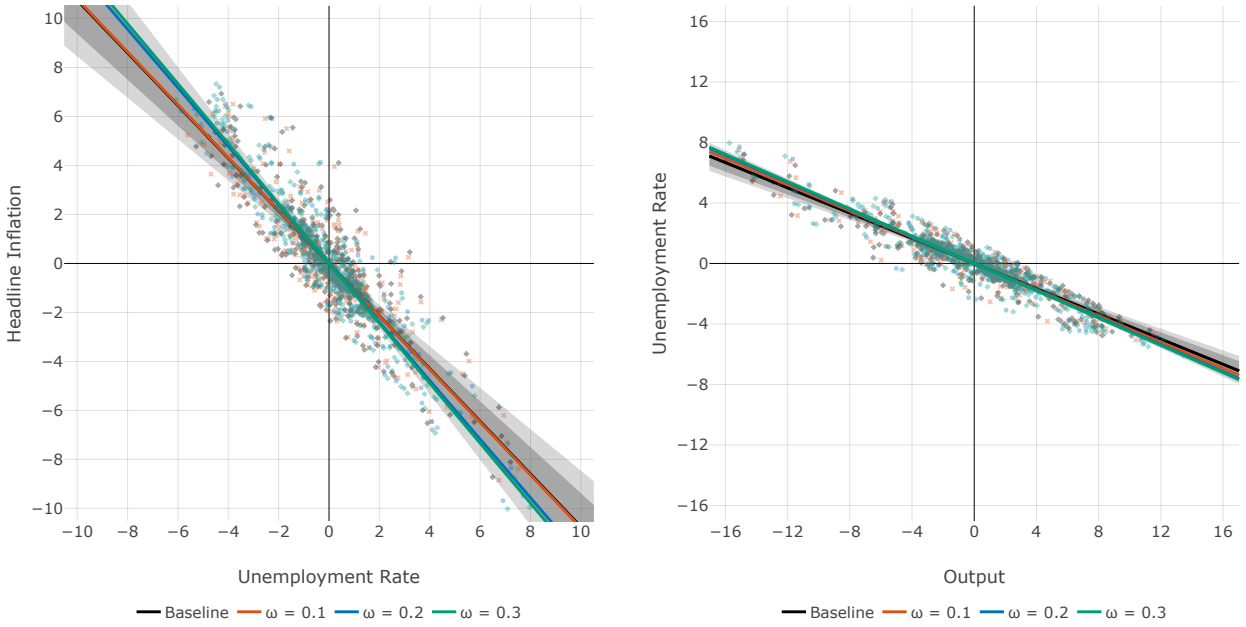
slopes, and both remain within the baseline posterior bands.[19]

5.3 The role of low-frequency movements in inflation

We conclude this section by asking what allows the model to recover a steep reduced-form Phillips curve. The model differs from a raw regression of inflation on unemployment in three

[19]The estimated correlations explain why these bounds are informative (Online Appendix Table C.1). The correlation between the cycle shock and the trend-inflation shock is negative, with posterior means of -0.18, -0.52 and -0.62 as ω rises from 0.1 to 0.3, well away from the prior mean of zero and with 68% intervals that exclude it throughout. The correlations with trend growth and with the NAIRU are smaller, negative for trend growth and positive for the NAIRU. The data are therefore informative about the sign of the trend-inflation correlation, but not about its size: its magnitude keeps rising with ω, so the prior is still binding at $\omega = 0.3$. We thus read this exercise as bounding the consequences of relaxing orthogonality rather than as estimating the correlation itself. Within the range the prior allows, which includes correlations well above 0.5 in absolute value, orthogonality does not drive our results.

Figure 11: Phillips curve and Okun's law with correlation between trends and cycle.



(a) Phillips curve

(b) Okun's law

Notes: In each panel: the baseline (orthogonal) model in black, with its 68/90% posterior credible band in grey, against the correlated trend–cycle specifications.

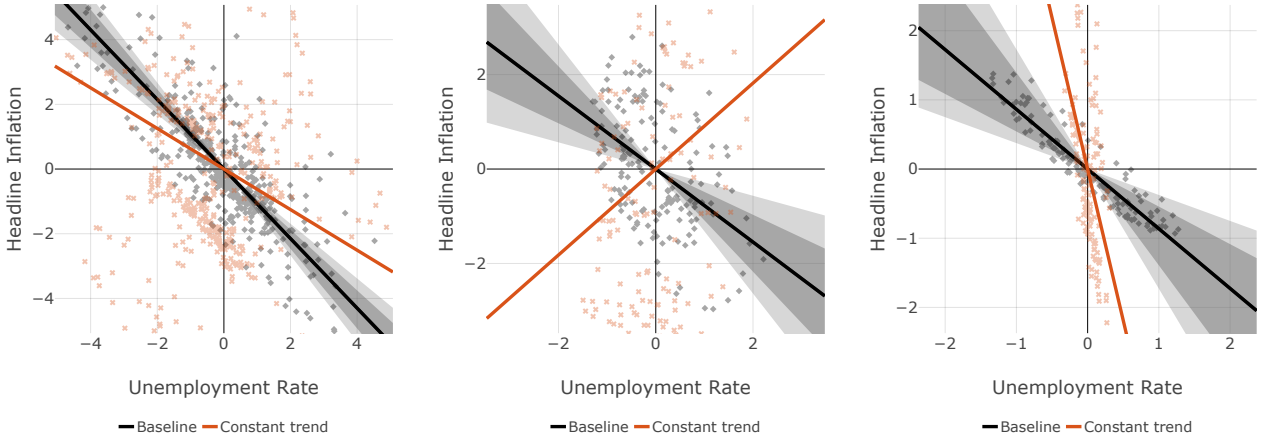
ways: it extracts a common cycle shared by all series, it allows for variable-specific idiosyncratic wedges, and it lets trend inflation, the NAIRU and potential output drift over time as random walks. The last of these turns out to be decisive. To show this, we re-estimate the model holding trend inflation constant, and compare the resulting reduced-form Phillips curve with the baseline (Figure 12).

Holding trend inflation constant overturns the result. Over the full century the curve flattens markedly. Over 1946–84 it turns positive.[20] This is the sample in which trend inflation moves most: with a constant trend, its rise through the Great Inflation, which coincided with rising unemployment, is read as cyclical, and the model recovers stagflation in place of a Phillips curve. Over 1985–2019 the relationship breaks down altogether. The cyclical component of unemployment almost disappears, while cyclical inflation absorbs the residual low-frequency movements, so that the fitted line is close to vertical and the slope is no longer meaningful. Without a time-varying trend, the common cycle therefore no longer isolates slack, and the inverse relationship between inflation and unemployment at business cycle frequency is lost. Okun's law, which links two real variables whose trends appear to be more stable, is far less exposed to this problem, consistent with its stability across the exercises above.[21]

[20] For 1946–84, the constant-trend model is the same specification used for this subsample in Section 5.1: the surveys are excluded and the cyclical components carry the same training-sample prior (see Section 3.2 and Online Appendix C.1). Only trend inflation changes, and is held constant.

[21] This finding is related to Hazell et al. (2022), who likewise argue that the aggregate evidence has been distorted by the low-frequency component of inflation. They control for it using state-level data, in which time fixed effects absorb long-run inflation expectations, and infer a small structural slope where we recover a steep reduced-form one. The two estimates are not in competition. Their structural slope is the response of

Figure 12: Phillips curve with constant trend inflation.



(a) 1900–2019 (b) 1946–1984 (c) 1985–2019

Notes: The reduced-form Phillips curve in the baseline model and with trend inflation held constant, by sample. In each panel: the baseline model in black, with its 68/90% posterior credible band in grey, against the constant-trend-inflation specification (orange).

6 Conclusions

The Phillips curve has been declared dead often enough that its obituaries may outnumber its estimates. This paper argues that reports of its death have been greatly exaggerated, and that what looked like a disappearance was in large part a measurement problem. Aggregate inflation and unemployment contain movements at frequencies other than that of the business cycle: permanent shifts in trend inflation and in the natural rate, as well as disturbances specific to each series. Once these are filtered out, a multivariate trend-cycle model recovers a steep, downward-sloping relationship between slack and prices across a century of monetary regimes.

Four results stand out. First, a single common cycle drives the joint fluctuations of output, unemployment, prices, expectations and the policy rate, and the signs of its loadings are those of a demand cycle. Second, at business cycle frequency the reduced-form Phillips curve is steep, with a slope of -1.08 over the full century. It is flatter, but of similar magnitude, in the two postwar subsamples, and no flatter in 1985–2019 than in 1946–84; and although it flattens in rolling windows that drop the early decades of the century, it remains firmly negative in every window. Third, Okun's law is in line with existing estimates and stable across subsamples and windows. Fourth, the common cycle accounts for most of the cyclical variation in the short-term interest rate, with a small idiosyncratic part confined largely to the Volcker disinflation; allowing policy to respond to the idiosyncratic component of inflation changes little. We read this as evidence that the cyclical fluctuations of the policy rate largely reflect systematic policy.

inflation to slack within a single period, holding expectations fixed. Ours is the contemporaneous comovement of inflation and slack as slack moves over the cycle: because the common cycle is persistent, today's gap carries information about the gap in the quarters ahead, and the reduced-form response cumulates what the structural one delivers period by period.

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